

THỰC HÀNH SOC

Bài 1 - GIỚI THIỆU

1. MỤC ĐÍCH

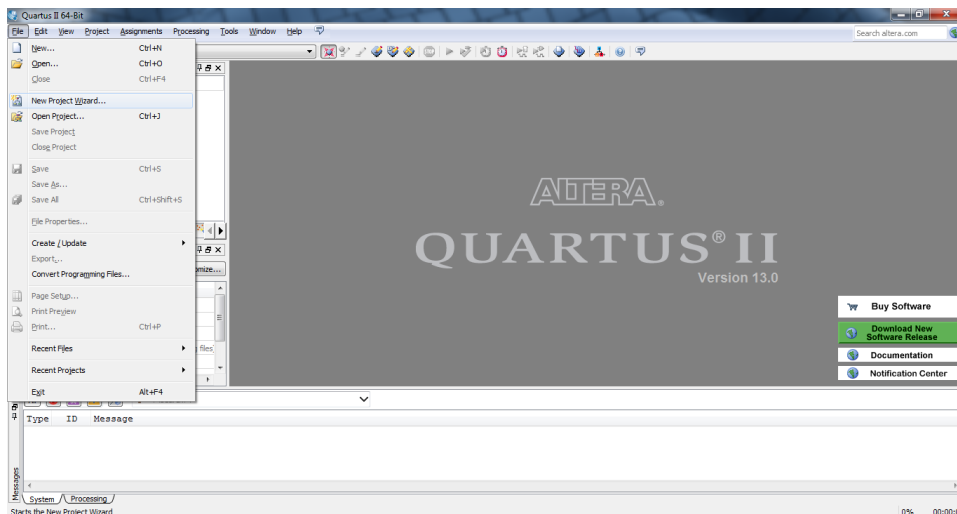
Thông qua bài thực hành này, sinh viên sẽ hiểu rõ

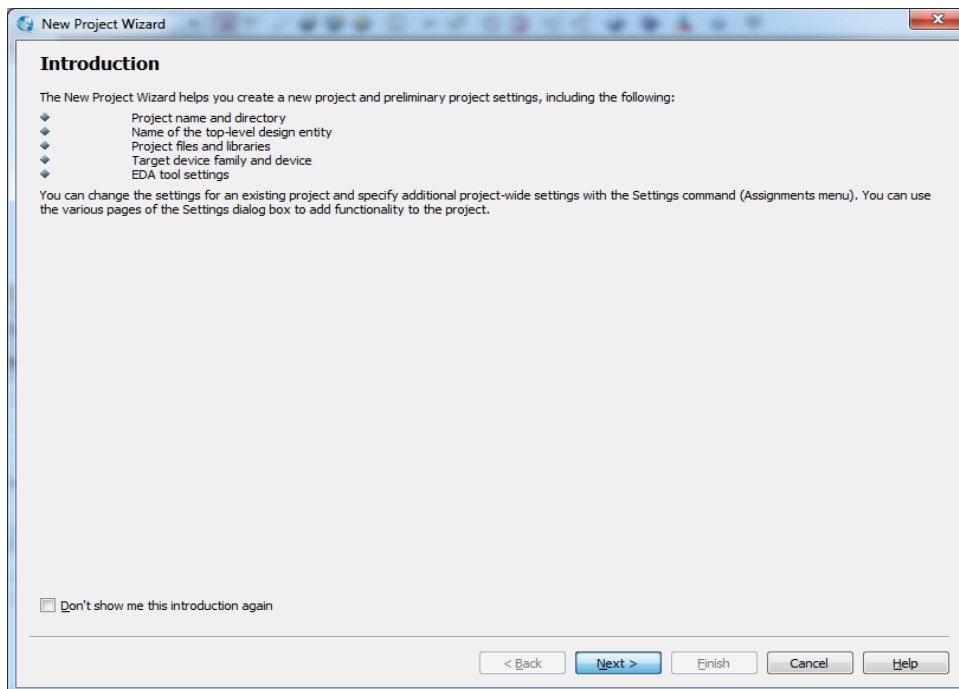
- o Cách thiết kế một máy tính hoàn chỉnh từ việc xây dựng phần cứng đến phần mềm.
- o Cách sử dụng công cụ Quartus II và Qsys để tạo phần cứng máy tính, tổng hợp và nạp xuống FPGA.
- o Cách sử dụng công cụ NIOS II để tổng hợp và nạp phần mềm xuống máy tính.

2. NỘI DUNG

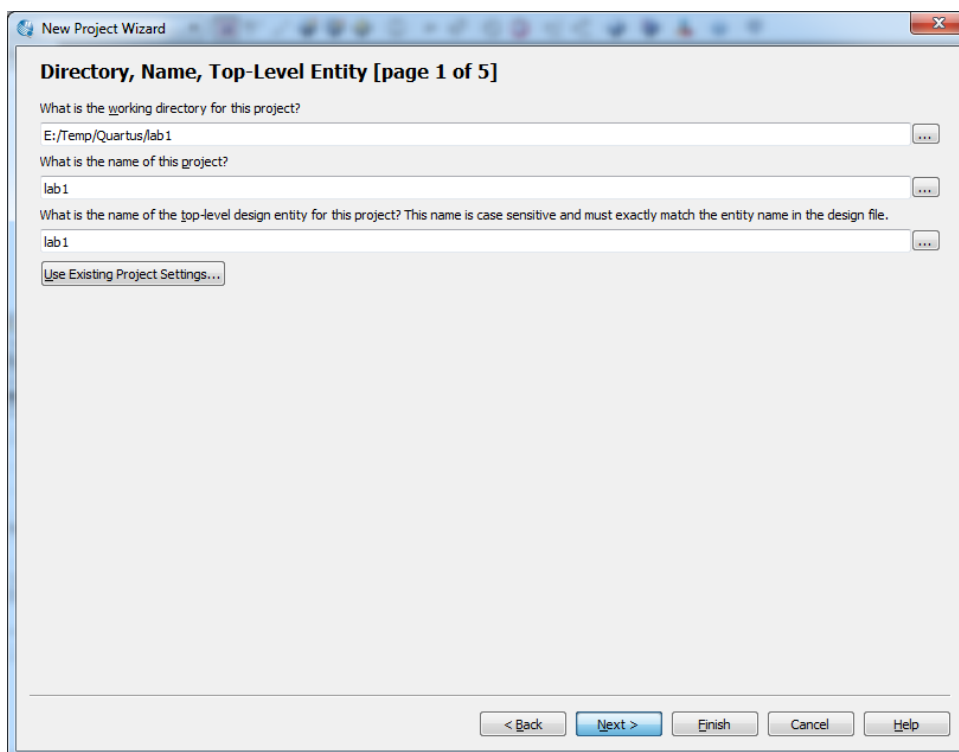
2.1. Hệ thống phần cứng

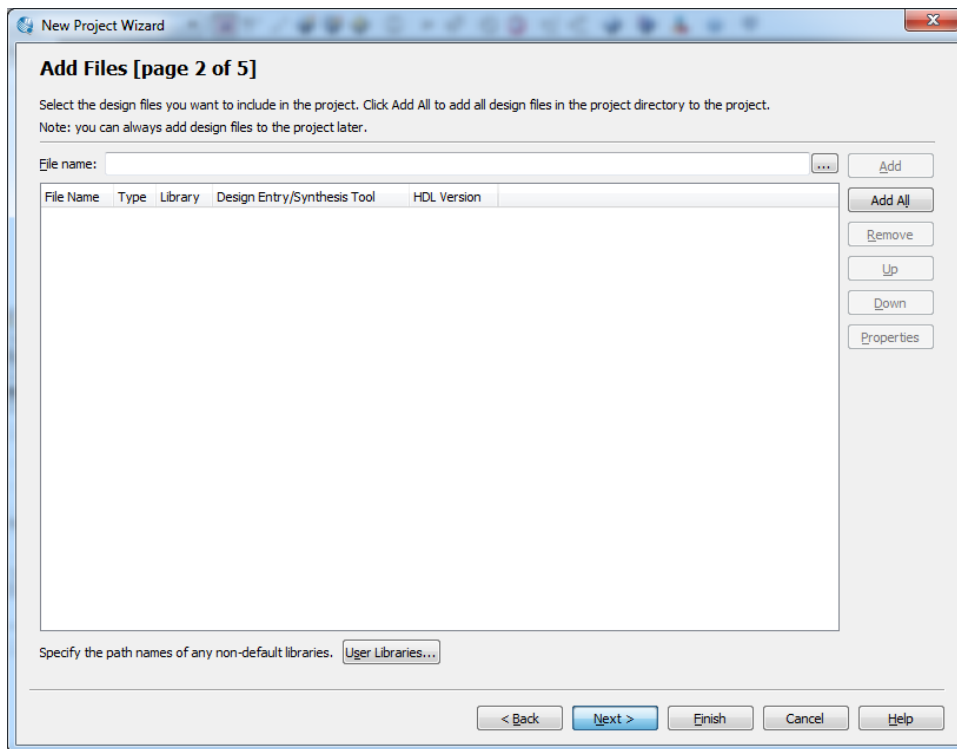
2.1.1. Tạo project Quartus



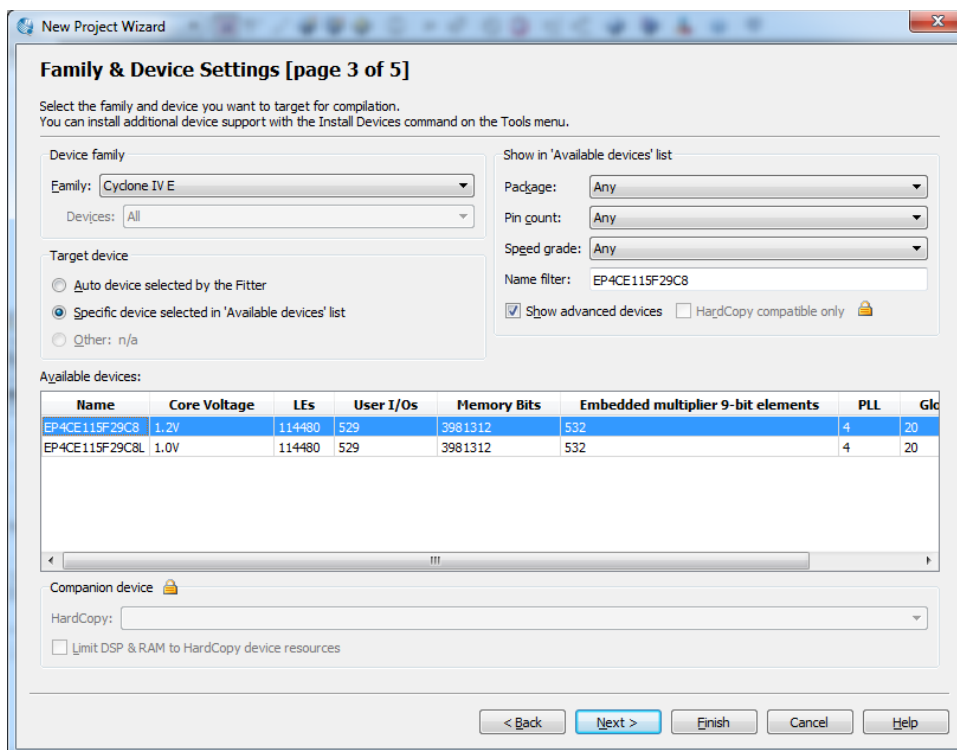


Đường dẫn thư mục chứa project không được có khoảng trắng





Nếu sử dụng board DE2-115, chọn Family là Cyclone IV E, device là EP4CE115F29C8



Nếu sử dụng board DE2, chọn Family là Cyclone II, device là EP2C35F672C6

New Project Wizard

Family & Device Settings [page 3 of 5]

Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.

Device family

Family: Cyclone II

Devices: All

Target device

☐ Auto device selected by the Fitter

☒ Specific device selected in 'Available devices' list

☐ Other: n/a

Show in 'Available devices' list

Package: Any

Pin count: Any

Speed grade: Any

Name filter: EP2C35F672C6

☒ Show advanced devices ☐ HardCopy compatible only

Available devices:

Name	Core Voltage	LEs	User I/Os	Memory Bits	Embedded multiplier 9-bit elements	PLL	Global
EP2C35F672C6	1.2V	33216	475	483840	70	4	16

Companion device

HardCopy:

☐ Limit DSP & RAM to HardCopy device resources

< Back Next > Finish Cancel Help

New Project Wizard

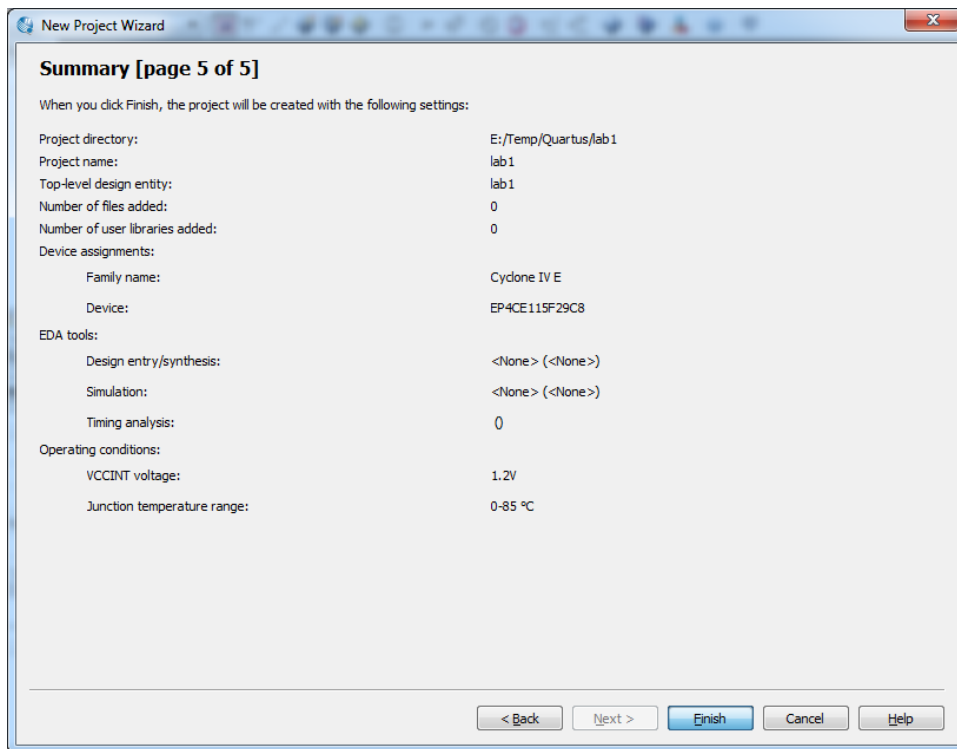
EDA Tool Settings [page 4 of 5]

Specify the other EDA tools used with the Quartus II software to develop your project.

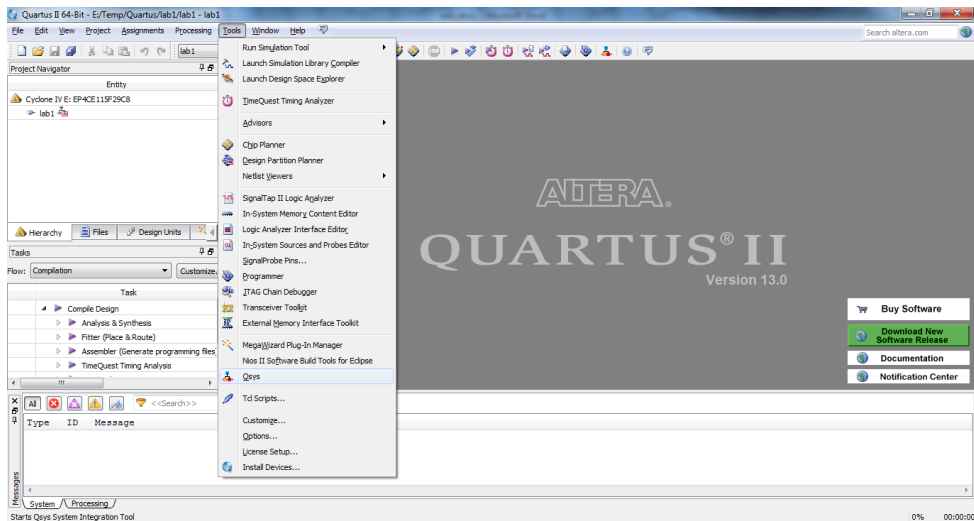
EDA tools:

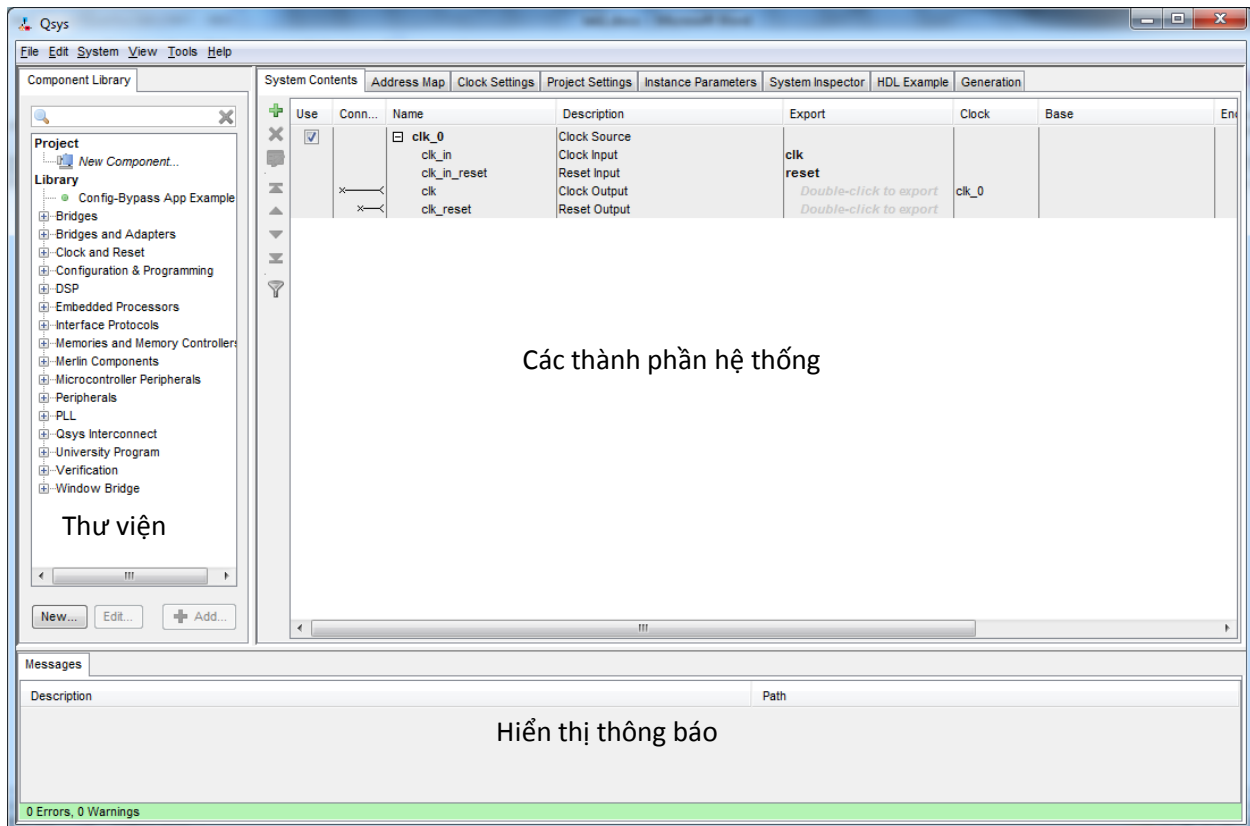
Tool Type	Tool Name	Format(s)	Run Tool Automatically
Design Entry/Synthesis	<None>	<None>	☐ Run this tool automatically to synthesize the current design
Simulation	<None>	<None>	☐ Run gate-level simulation automatically after compilation
Formal Verification	<None>	<None>	
Board-Level	Timing	<None>	
	Symbol	<None>	
	Signal Integrity	<None>	
	Boundary Scan	<None>	

< Back Next > Finish Cancel Help

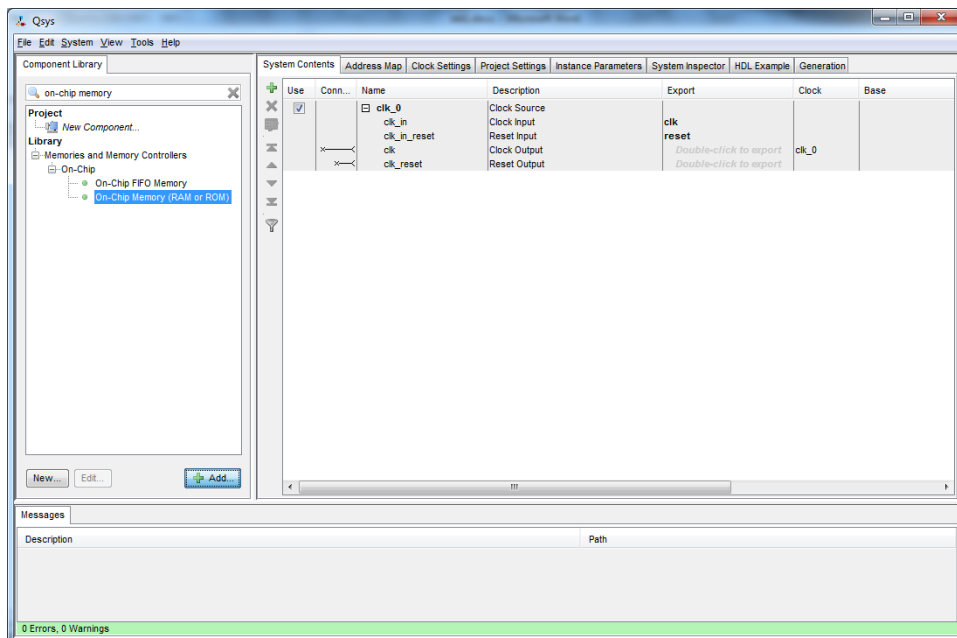


2.1.2. Xây dựng hệ thống SoPC sử dụng Qsys

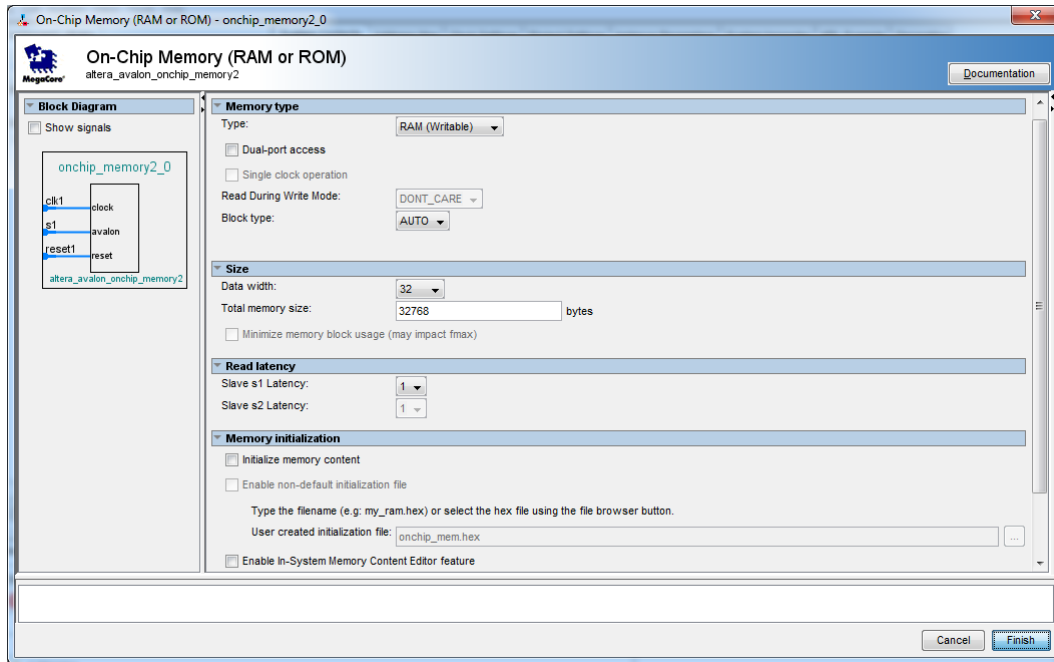




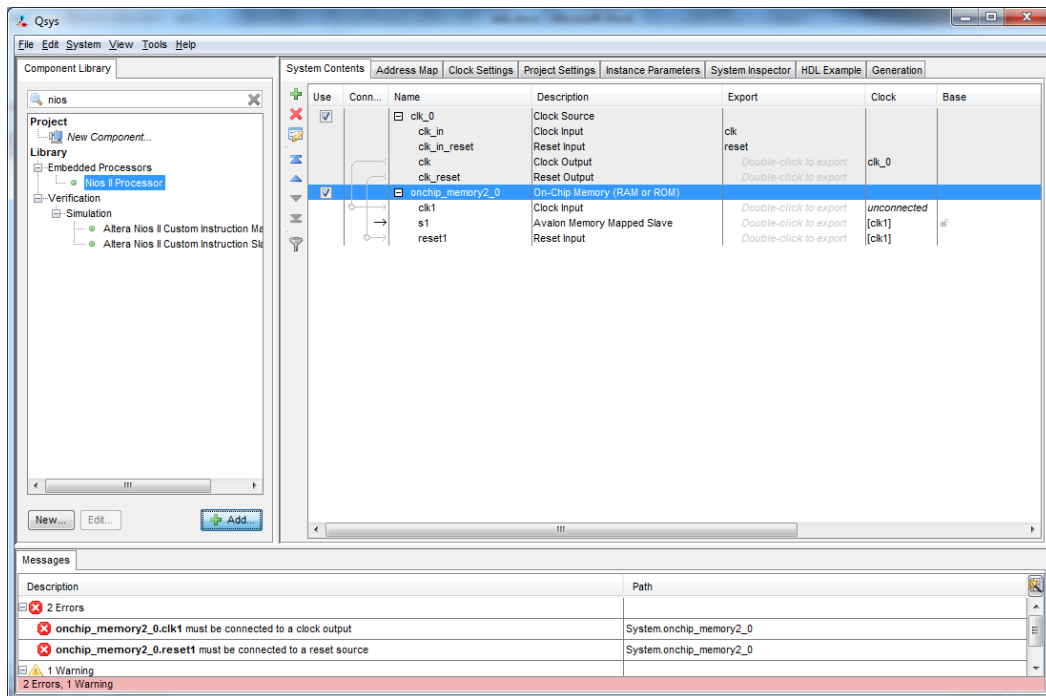
Trong cửa sổ thư viện, gõ vào ô tìm kiếm “On-chip memory”, chọn “on-chip memory (RAM or ROM)”, click “Add”



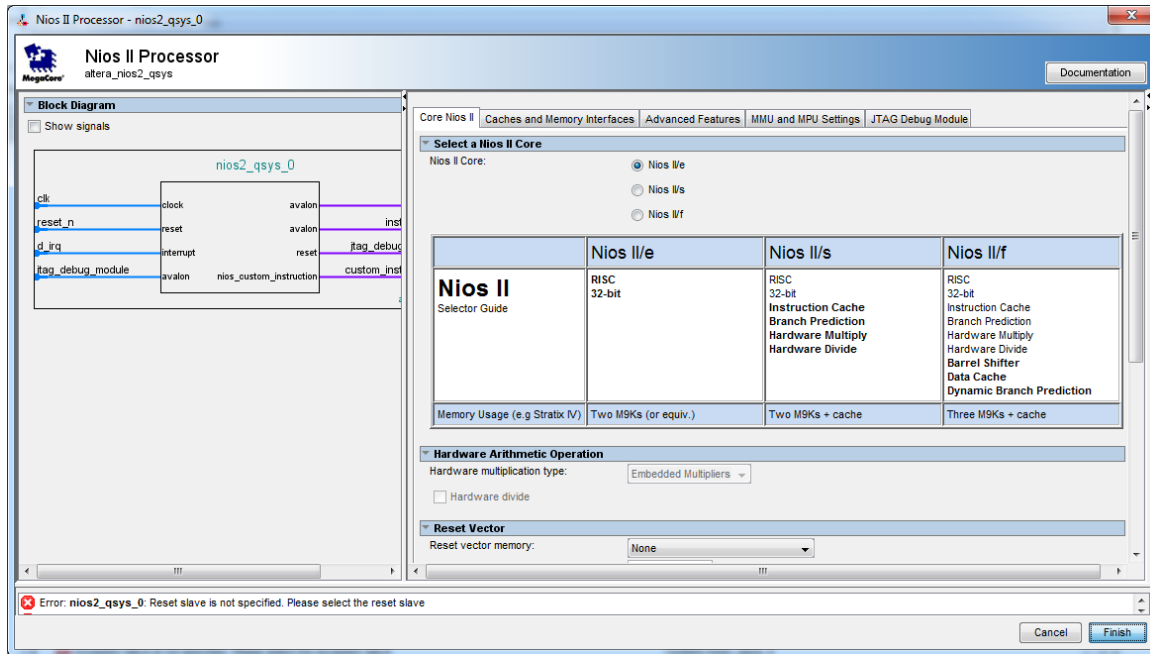
Cấu hình như bên dưới, click “Finish”



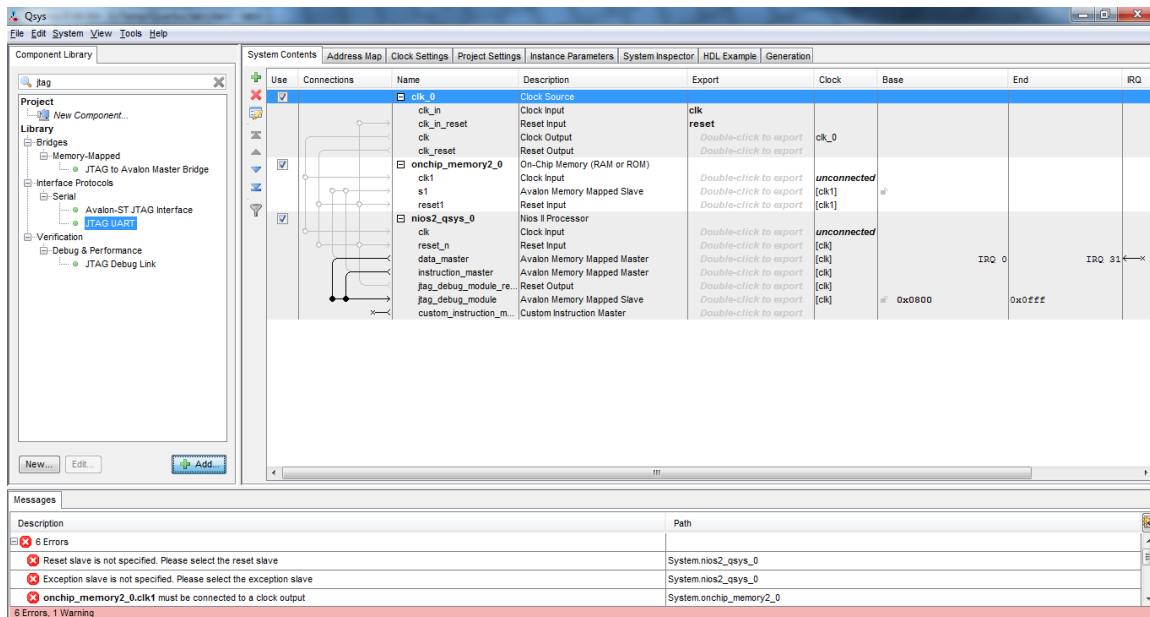
Tìm “Nios II processor” và thêm vào hệ thống



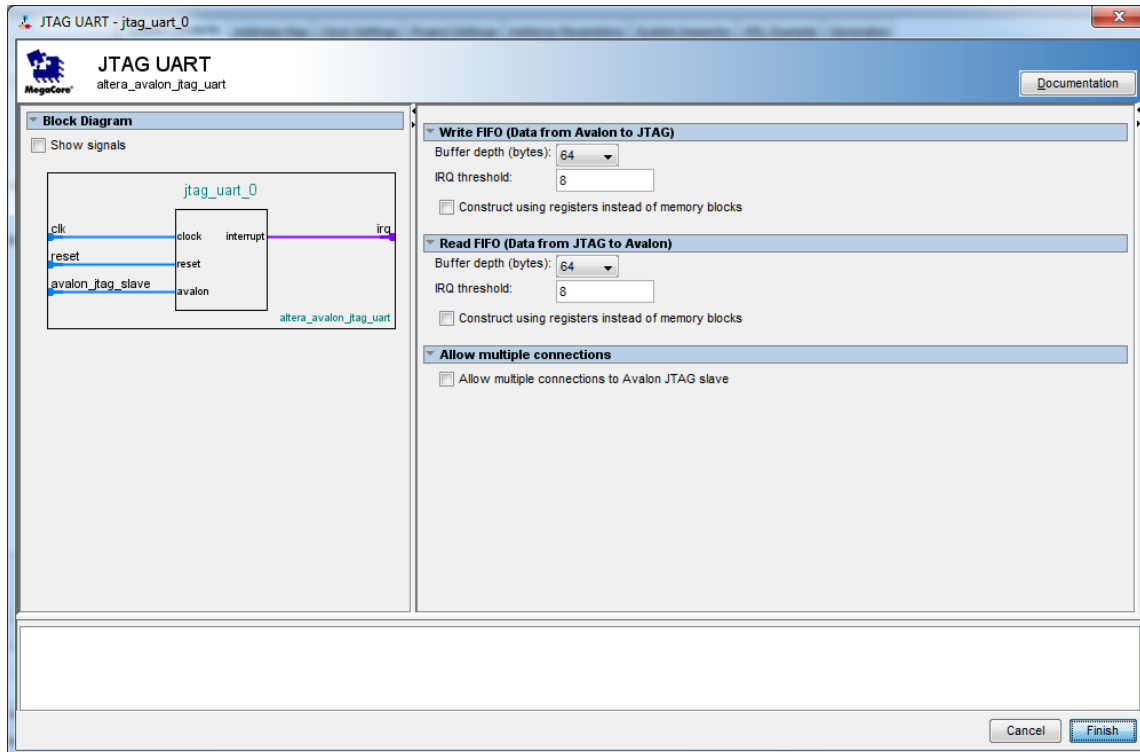
Ở tab “Core Nios II”, chọn “Nios II/e”, click “Finish”



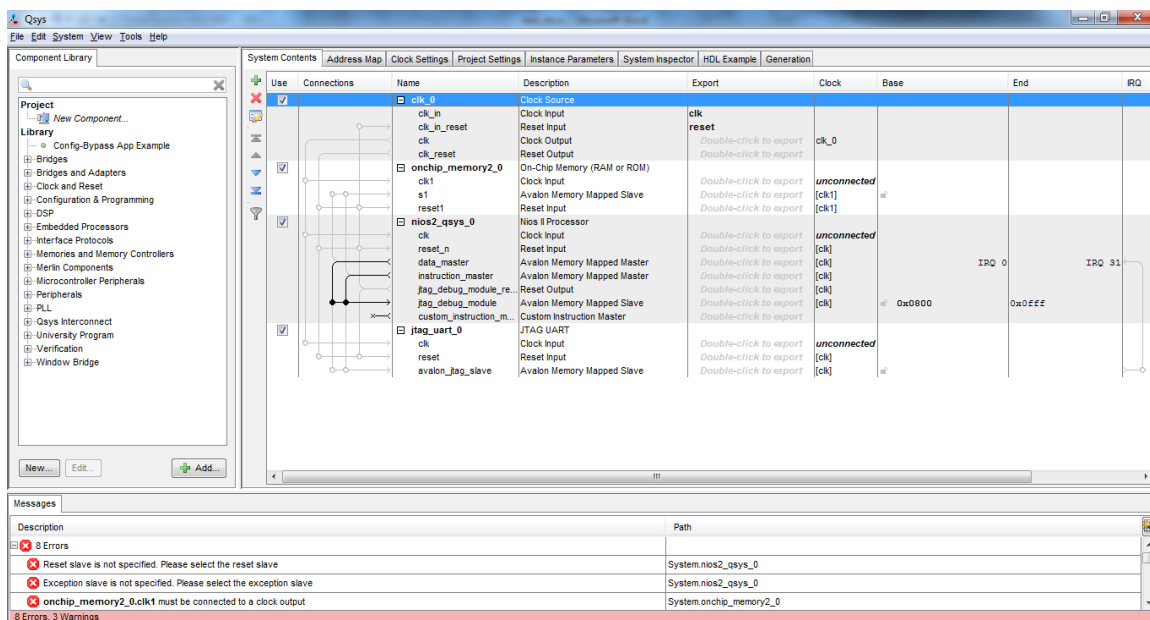
Tìm kiếm “JTAG UART” và thêm vào hệ thống



Click “Finish”



Hệ thống hoàn chỉnh như hình bên dưới



Kết nối tín hiệu “clk” từ module “clk_0” đến các module khác

The screenshot shows the Qsys interface with the 'System Contents' tab selected. The 'Connections' column shows the following connections:

- clk_0** (Clock Source) is connected to **clk_in** (Clock Input) of **clk_0**.
- clk_in_reset** (Reset Input) is connected to **reset** (Reset Input) of **clk_0**.
- clk_reset** (Reset Output) is connected to **reset** (Reset Input) of **onchip_memory2_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **onchip_memory2_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **nios2_qsys_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **jtag_uart_0**.

The 'Messages' pane at the bottom shows 5 Errors and 3 Warnings:

- Reset slave is not specified. Please select the reset slave. Path: System.nios2_qsys_0
- Exception slave is not specified. Please select the exception slave. Path: System.nios2_qsys_0
- onchip_memory2_0.reset must be connected to a reset source. Path: System.onchip_memory2_0

Kết nối tín hiệu “clk_reset” từ module “clk_0” đến các module khác

The screenshot shows the Qsys interface with the 'System Contents' tab selected. The 'Connections' column shows the following connections:

- clk_0** (Clock Source) is connected to **clk_in** (Clock Input) of **clk_0**.
- clk_in_reset** (Reset Input) is connected to **reset** (Reset Input) of **clk_0**.
- clk_reset** (Reset Output) is connected to **reset** (Reset Input) of **onchip_memory2_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **onchip_memory2_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **nios2_qsys_0**.
- clk_0** (Clock Output) is connected to **clk** (Clock Input) of **jtag_uart_0**.

The 'Messages' pane at the bottom shows 2 Errors and 3 Warnings:

- Reset slave is not specified. Please select the reset slave. Path: System.nios2_qsys_0
- Exception slave is not specified. Please select the exception slave. Path: System.nios2_qsys_0

Kết nối interface “data master” từ module “nios2_qsys_0” đến các interface khác

The screenshot shows the Qsys software interface with the 'System Contents' tab selected. The 'Connections' pane on the left shows a tree view of the system components. The 'Data Master' interface is highlighted in blue. The 'Messages' pane at the bottom shows 6 errors and 1 warning.

Name	Description	Export	Clock	Base	End	IRQ
clk_0	Clock Source	clk	clk_0			
clk_in	Clock Input	clk	clk_0			
clk_in_reset	Reset Input	clk	clk_0			
clk_reset	Reset Output	clk	clk_0			
onchip_memory2_0	On-Chip Memory (RAM or ROM)	clk	clk_0			
clk1	Clock Input	clk1	clk_0			
reset1	Reset Input	clk1	clk_0			
nios2_qsys_0	Nios II Processor	clk	clk_0			
reset_n	Reset Input	clk	clk_0			
data_master	Avalon Memory Mapped Master	clk	clk_0			
instruction_master	Avalon Memory Mapped Master	clk	clk_0			
jtag_debug_module_re	Reset Output	clk	clk_0			
jtag_debug_module	Avalon Memory Mapped Slave	clk	clk_0			
custom_instruction_m...	Custom Instruction Master	clk	clk_0			
jtag_uart_0	JTAG UART	clk	clk_0			
clk	Clock Input	clk	clk_0			
reset	Reset Input	clk	clk_0			
avalon_jtag_slave	Avalon Memory Mapped Slave	clk	clk_0			

Messages:

- Reset slave is not specified. Please select the reset slave
- Exception slave is not specified. Please select the exception slave
- onchip_memory2_0.s1 (0x0..0x7fff) overlaps jtag_uart_0.avalon_jtag_slave (0x0..0x7)

Kết nối interface “instruction master” từ module “nios2_qsys_0” đến các interface khác

The screenshot shows the Qsys software interface with the 'System Contents' tab selected. The 'Connections' pane on the left shows a tree view of the system components. The 'Instruction Master' interface is highlighted in blue. The 'Messages' pane at the bottom shows 10 errors and 1 warning.

Name	Description	Export	Clock	Base	End	IRQ
clk_0	Clock Source	clk	clk_0			
clk_in	Clock Input	clk	clk_0			
clk_in_reset	Reset Input	clk	clk_0			
clk_reset	Reset Output	clk	clk_0			
onchip_memory2_0	On-Chip Memory (RAM or ROM)	clk	clk_0			
clk1	Clock Input	clk1	clk_0			
reset1	Reset Input	clk1	clk_0			
nios2_qsys_0	Nios II Processor	clk	clk_0			
reset_n	Reset Input	clk	clk_0			
data_master	Avalon Memory Mapped Master	clk	clk_0			
instruction_master	Avalon Memory Mapped Master	clk	clk_0			
jtag_debug_module_re	Reset Output	clk	clk_0			
jtag_debug_module	Avalon Memory Mapped Slave	clk	clk_0			
custom_instruction_m...	Custom Instruction Master	clk	clk_0			
jtag_uart_0	JTAG UART	clk	clk_0			
clk	Clock Input	clk	clk_0			
reset	Reset Input	clk	clk_0			
avalon_jtag_slave	Avalon Memory Mapped Slave	clk	clk_0			

Messages:

- Reset slave is not specified. Please select the reset slave
- Exception slave is not specified. Please select the exception slave
- onchip_memory2_0.s1 (0x0..0x7fff) overlaps jtag_uart_0.avalon_jtag_slave (0x0..0x7)

Kết nối interface “jtag_debug_module_reset” từ module “nios2_qsys_0” đến các interface khác

Qsys - nios_sys.qsys* (E:\Temp\Quartus\lab1\nios_sys.qsys)

File Edit System View Tools Help

Component Library

Project: New Component...

Library: Config-Bypass App Example, Bridges, Bridges and Adapters, Clock and Reset, Configuration & Programming, DSP, Embedded Processors, Interface Protocols, Memories and Memory Controllers, Merlin Components, Microcontroller Peripherals, Peripherals, PLL, Qsys Interconnect, University Program, Verification, Window Bridge

System Contents: Address Map, Clock Settings, Project Settings, Instance Parameters, System Inspector, HDL Example, Generation

Use	Connections	Name	Description	Export	Clock	Base	End	IRQ	Opcode Nat
☒		clk_0	Clock Source	clk	clk_0				
☒		clk_in	Clock Input	reset					
☒		clk_in_reset	Reset Input	Double-click to export					
☒		clk_reset	Reset Output	Double-click to export					
☒		onchip_memory2_0	On-Chip Memory (RAM or ROM)						
☒		clk1	Clock Input	Double-click to export	clk_0				
☒		s1	Avalon Memory Mapped Slave	Double-click to export	[clk1]	# 0x0000_8000	0x0000_ffff		
☒		reset1	Reset Input	Double-click to export					
☒		nios2_qsys_0	Nios II Processor						
☒		clk	Clock Input	Double-click to export	clk_0				
☒		reset_n	Reset Input	Double-click to export	[clk]				
☒		data_master	Avalon Memory Mapped Master	Double-click to export	[clk]			IRQ 0	IRQ 31
☒		instruction_master	Avalon Memory Mapped Master	Double-click to export	[clk]				
☒		jtag_debug_module_reset	Reset Output	Double-click to export	[clk]				
☒		jtag_debug_module	Avalon Memory Mapped Slave	Double-click to export	[clk]	# 0x0001_0800	0x0001_0fff		
☒		custom_instruction_master	Custom Instruction Master	Double-click to export	[clk]				
☒		jtag_uart_0	JTAG UART						
☒		clk	Clock Input	Double-click to export	clk_0				
☒		reset	Reset Input	Double-click to export	[clk]				
☒		avalon_jtag_slave	Avalon Memory Mapped Slave	Double-click to export	[clk]	# 0x0001_1008	0x0001_100f		

Messages

Description: Path

1 Warning: Interrupt sender jtag_uart_0.irq is not connected to an interrupt receiver. Path: System.jtag_uart_0

0 Errors, 1 Warning

Kết nối tín hiệu “irq” từ module “jtag_uart_0” đến module “nios_qsys_0”

Qsys - nios_sys.qsys* (E:\Temp\Quartus\lab1\nios_sys.qsys)

File Edit System View Tools Help

Component Library

Project: New Component...

Library: Config-Bypass App Example, Bridges, Bridges and Adapters, Clock and Reset, Configuration & Programming, DSP, Embedded Processors, Interface Protocols, Memories and Memory Controllers, Merlin Components, Microcontroller Peripherals, Peripherals, PLL, Qsys Interconnect, University Program, Verification, Window Bridge

System Contents: Address Map, Clock Settings, Project Settings, Instance Parameters, System Inspector, HDL Example, Generation

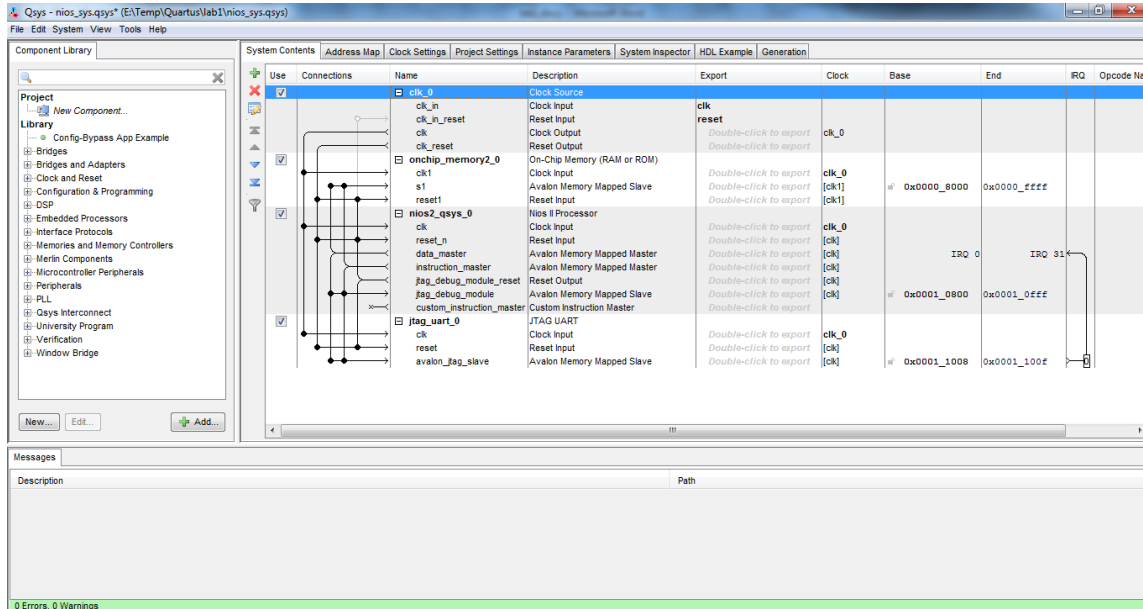
Use	Connections	Name	Description	Export	Clock	Base	End	IRQ	Opcode Nat
☒		clk_0	Clock Source	clk	clk_0				
☒		clk_in	Clock Input	reset					
☒		clk_in_reset	Reset Input	Double-click to export					
☒		clk_reset	Reset Output	Double-click to export					
☒		onchip_memory2_0	On-Chip Memory (RAM or ROM)						
☒		clk1	Clock Input	Double-click to export	clk_0				
☒		s1	Avalon Memory Mapped Slave	Double-click to export	[clk1]	# 0x0000_8000	0x0000_ffff		
☒		reset1	Reset Input	Double-click to export					
☒		nios2_qsys_0	Nios II Processor						
☒		clk	Clock Input	Double-click to export	clk_0				
☒		reset_n	Reset Input	Double-click to export	[clk]				
☒		data_master	Avalon Memory Mapped Master	Double-click to export	[clk]			IRQ 0	IRQ 31
☒		instruction_master	Avalon Memory Mapped Master	Double-click to export	[clk]				
☒		jtag_debug_module_reset	Reset Output	Double-click to export	[clk]				
☒		jtag_debug_module	Avalon Memory Mapped Slave	Double-click to export	[clk]	# 0x0001_0800	0x0001_0fff		
☒		custom_instruction_master	Custom Instruction Master	Double-click to export	[clk]				
☒		jtag_uart_0	JTAG UART						
☒		clk	Clock Input	Double-click to export	clk_0				
☒		reset	Reset Input	Double-click to export	[clk]				
☒		avalon_jtag_slave	Avalon Memory Mapped Slave	Double-click to export	[clk]	# 0x0001_1008	0x0001_100f		

Messages

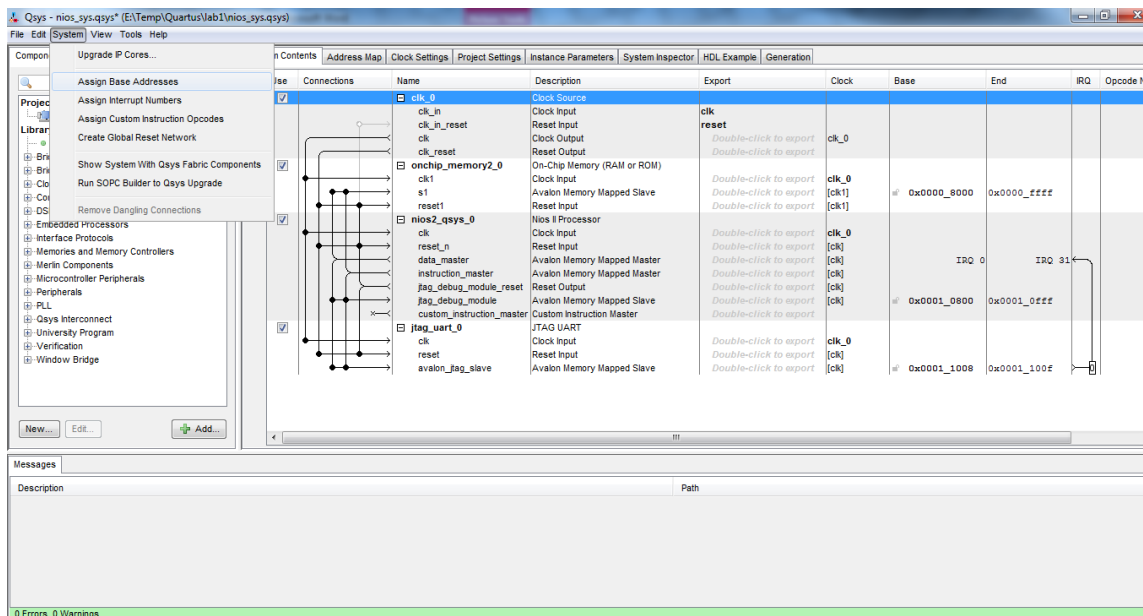
Description: Path

0 Errors, 0 Warnings

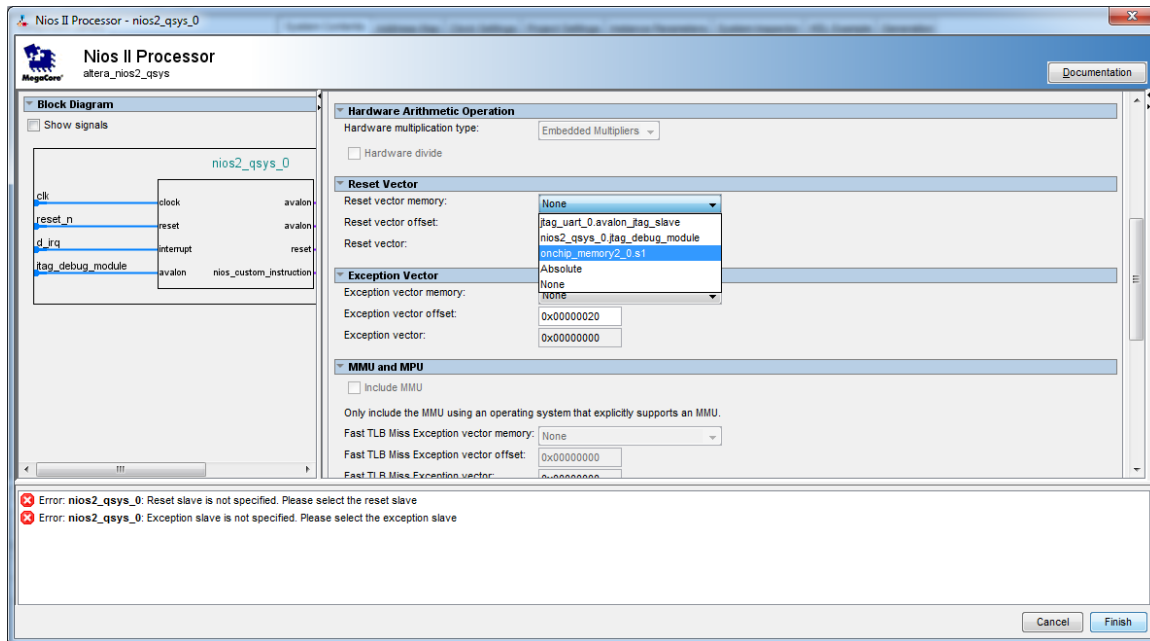
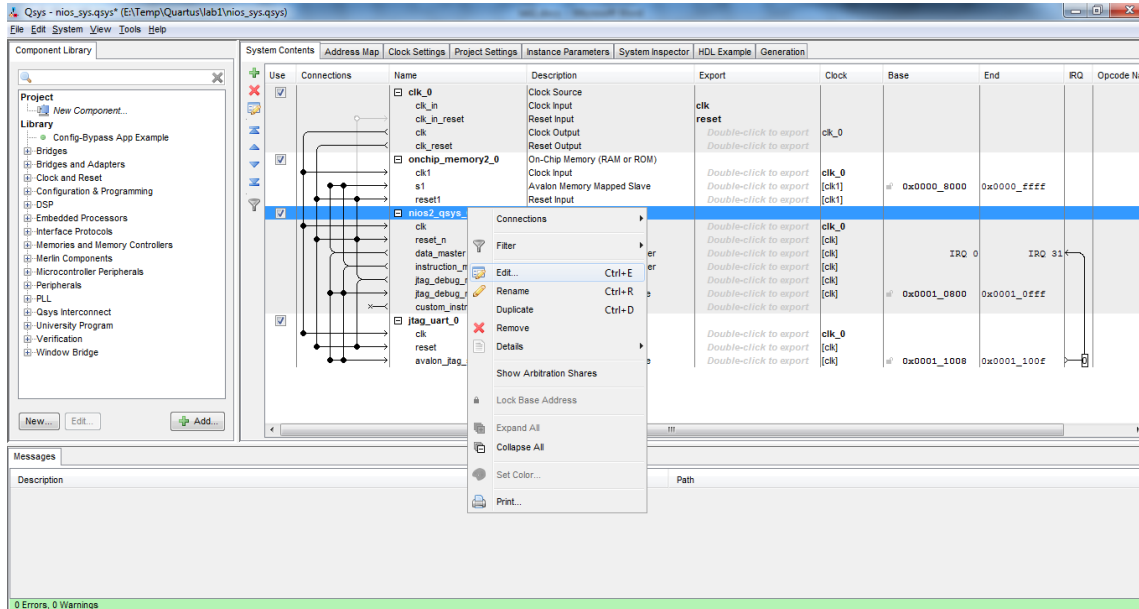
Sau khi kết nối xong sẽ được hệ thống như bên dưới

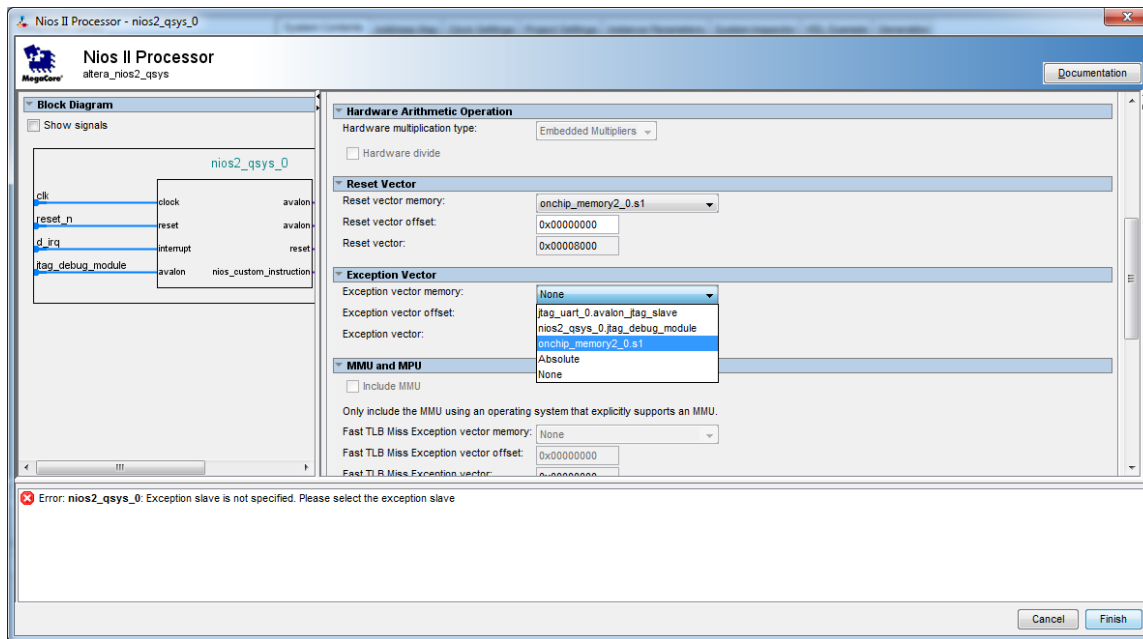


Gán địa chỉ cho các module



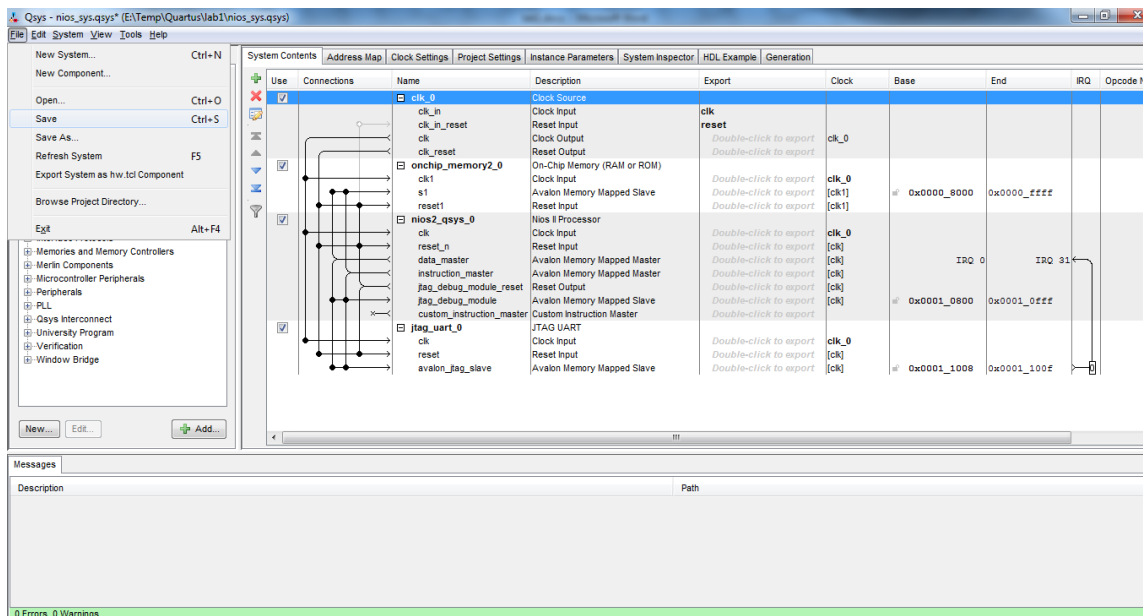
Sửa lại cấu hình của module “nios_qsys_0”

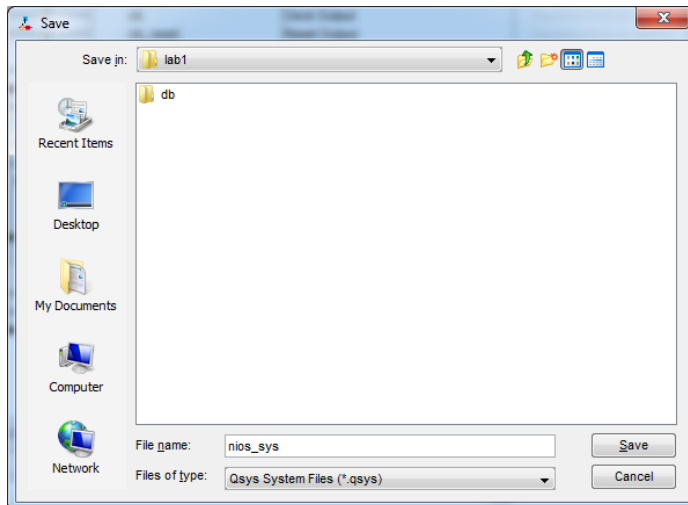




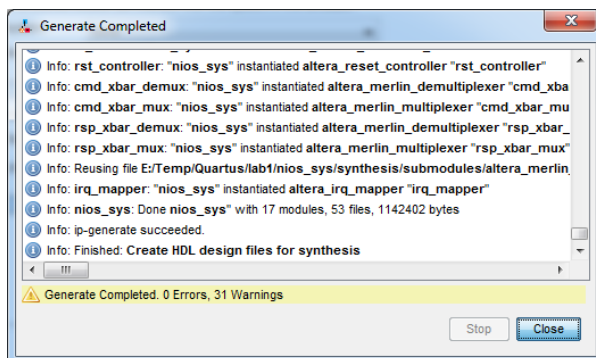
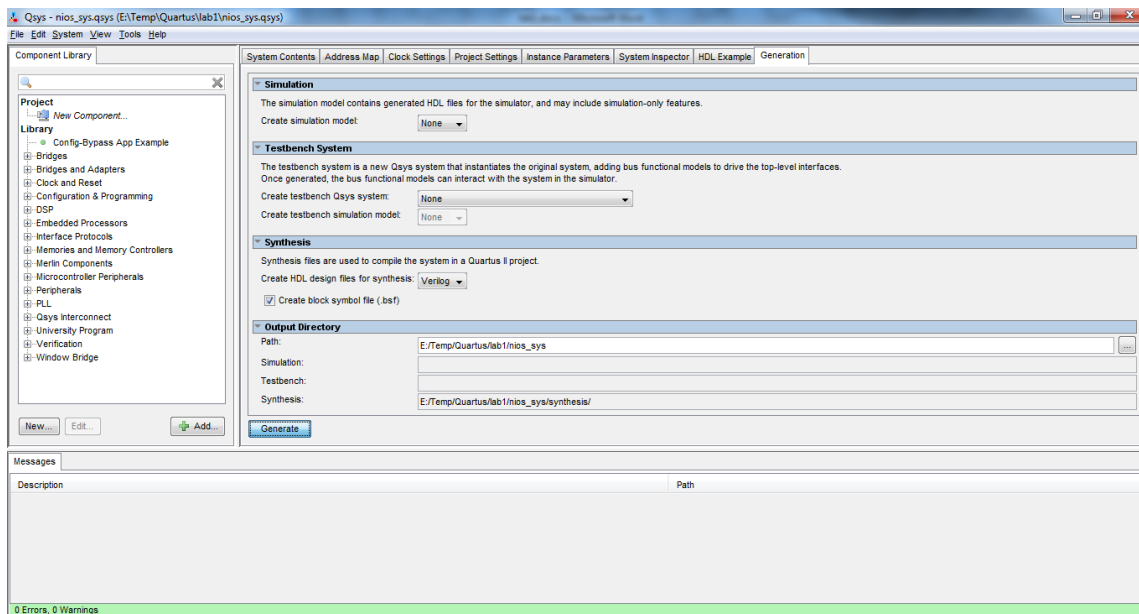
Click “Finish”

Hệ thống phần cứng đã hoàn thành, không còn thông báo lỗi. Save lại hệ thống với tên “nios_sys”

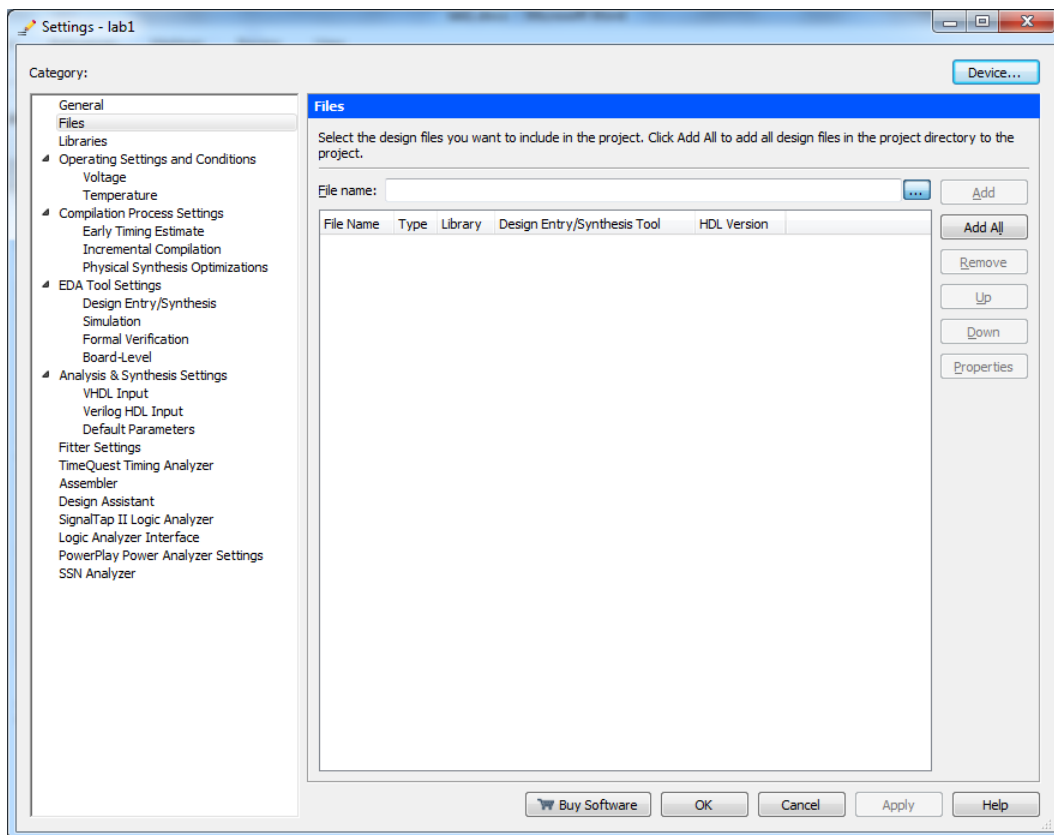
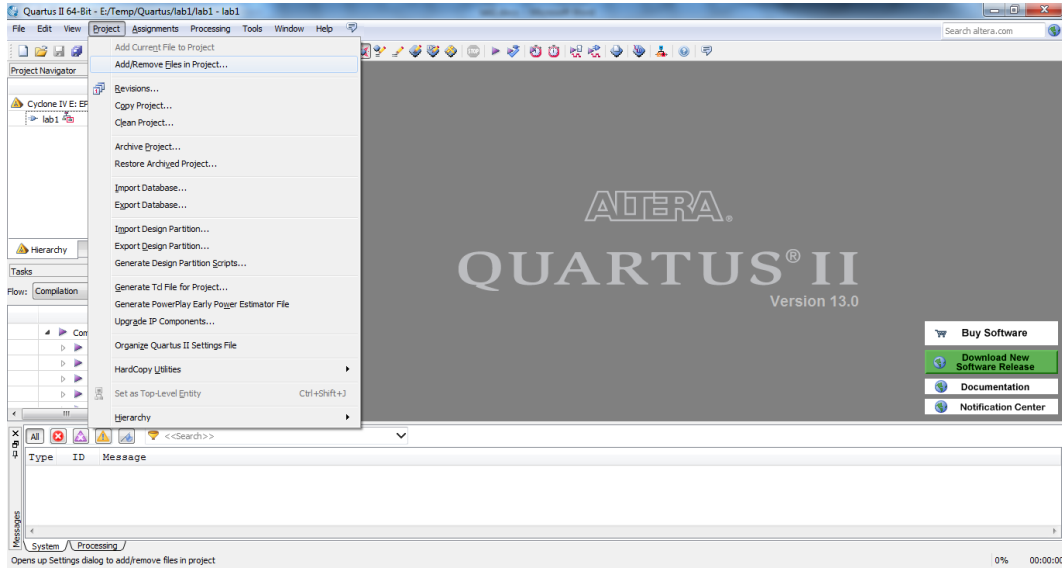




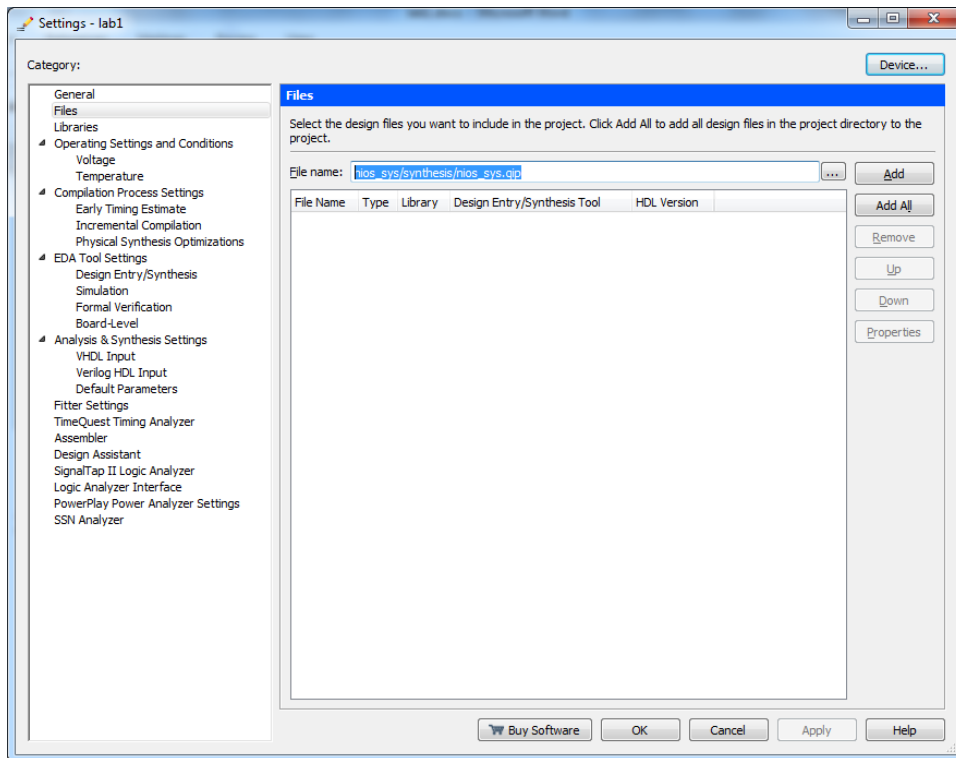
Chuyển sang tab “Generation”, click “Generate”



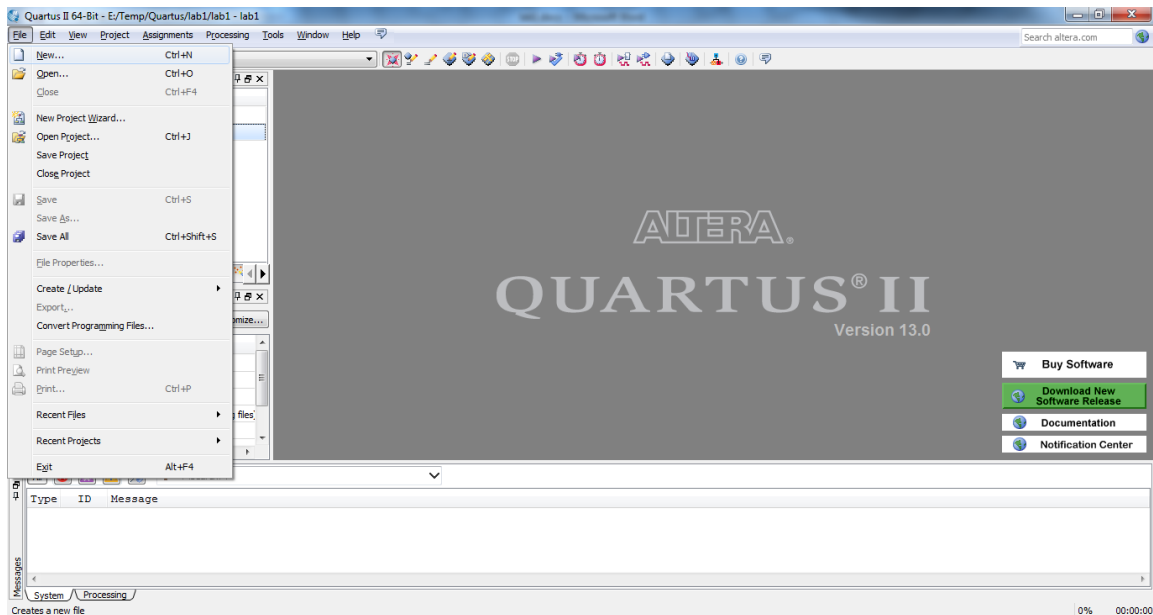
2.1.3. Tích hợp hệ thống vào project Quartus

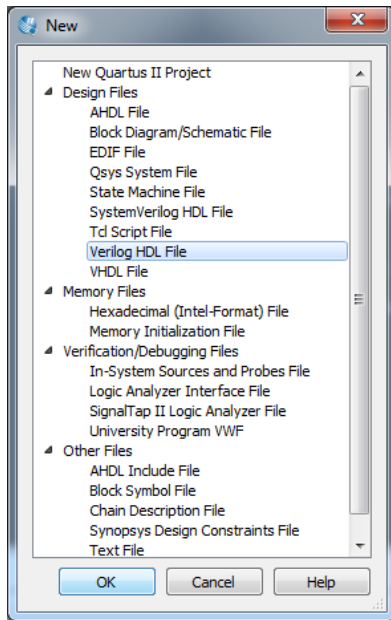


Gõ đường dẫn “nios_sys/synthesis/nios_sys.qip” vào ô “File name” và click “Add”, click “OK”

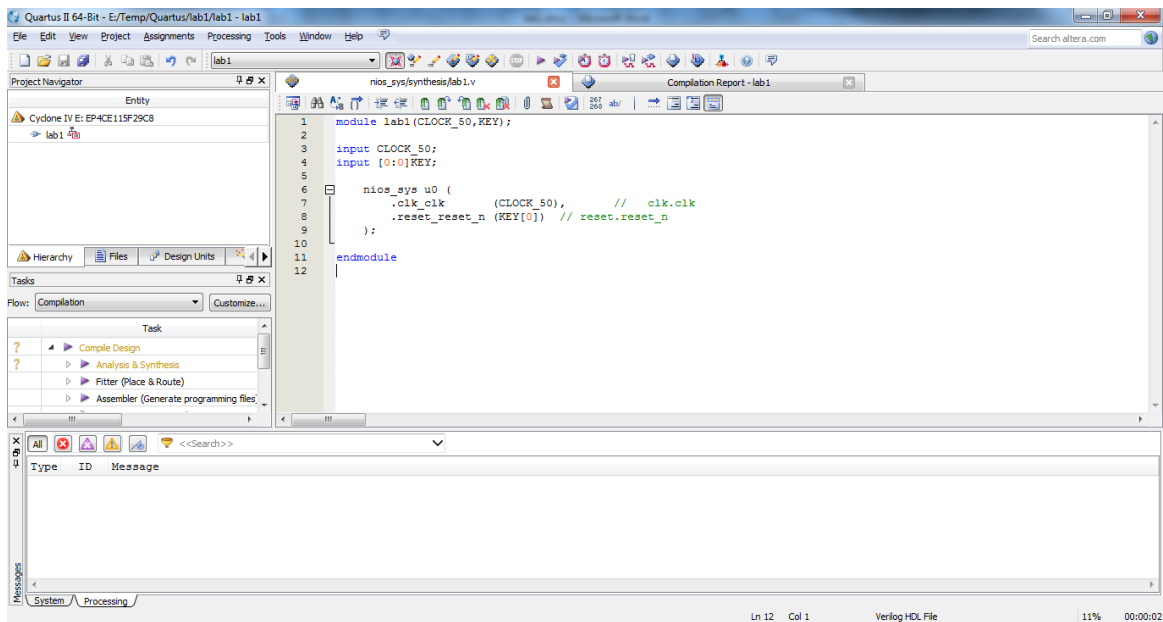


Tạo file top module, đặt tên là “lab1.v”



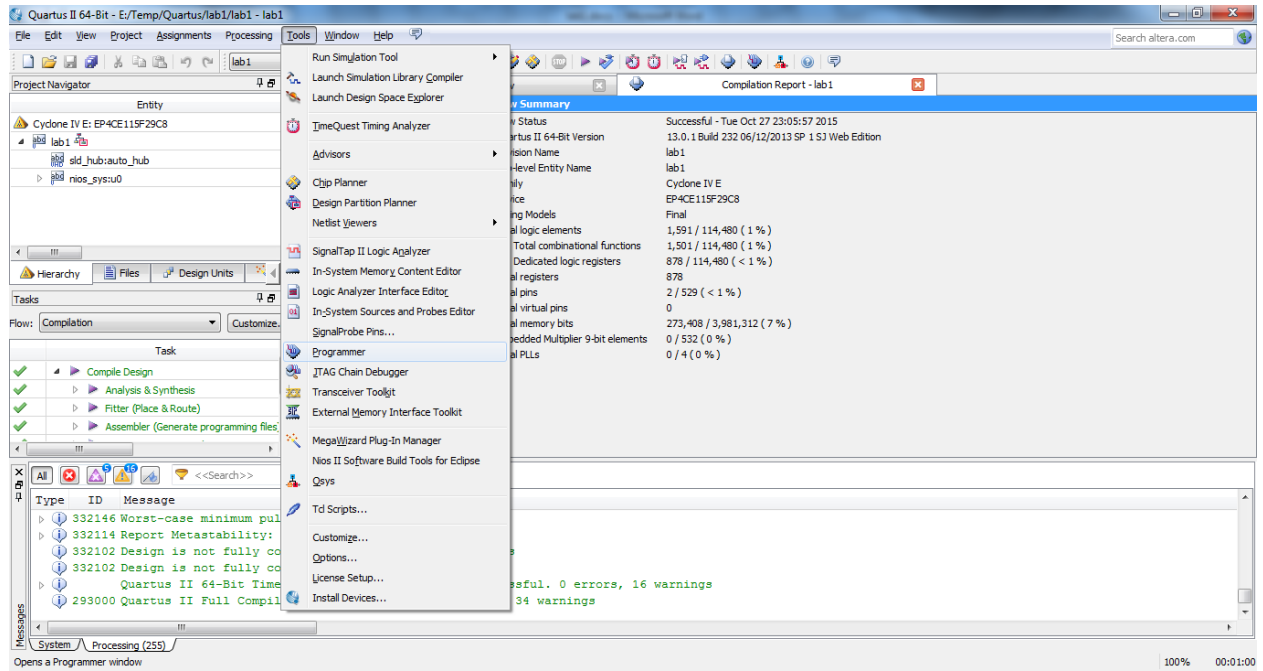


Nội dung của file “lab1.v” như hình bên dưới

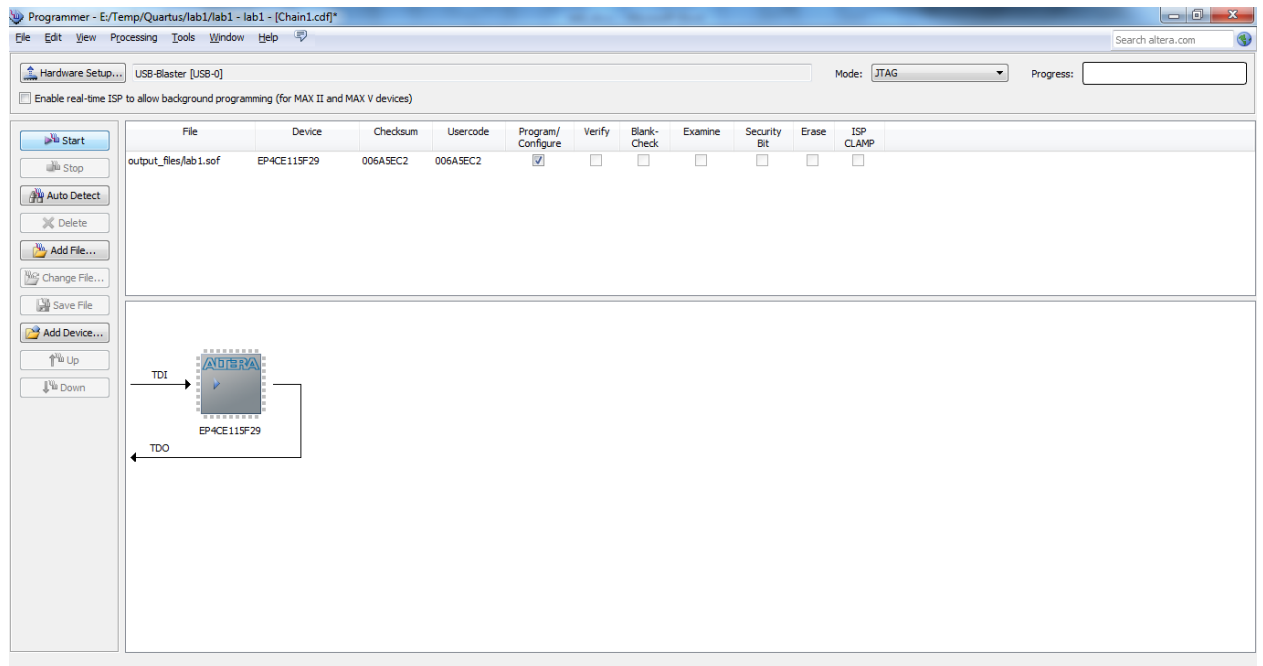


Gán pin cho các tín hiệu (Assignment > Import assignments...) và build project Quartus.

Download hệ thống phần cứng xuống board

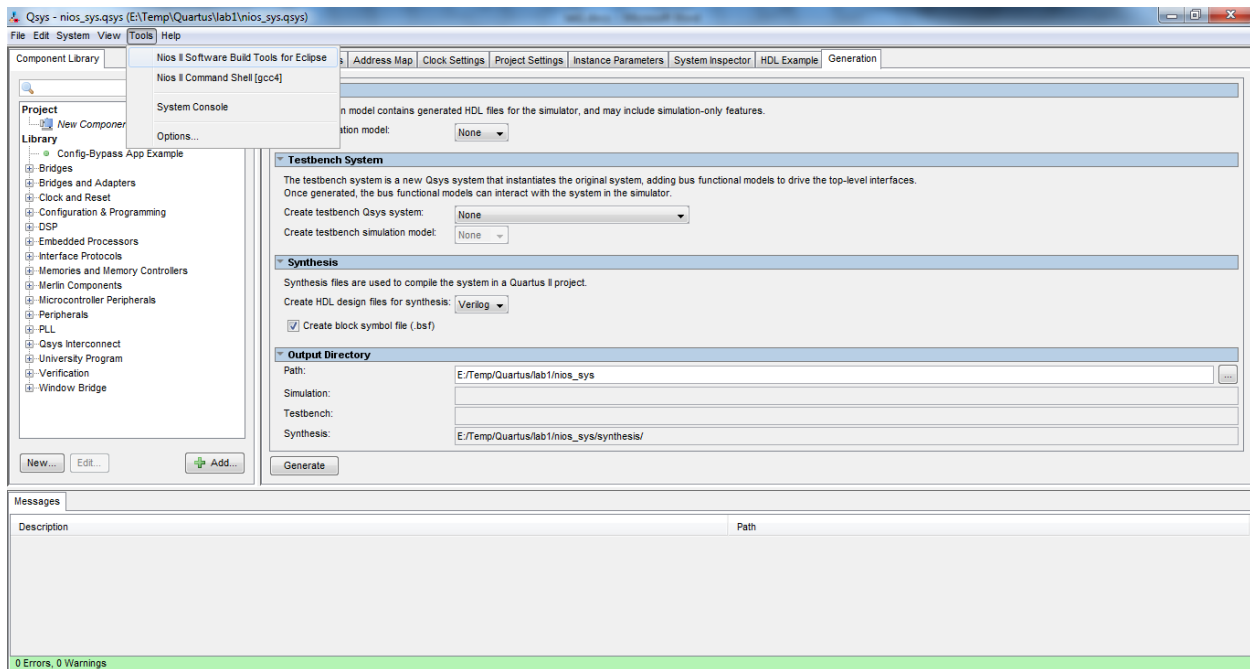


Click “Start”

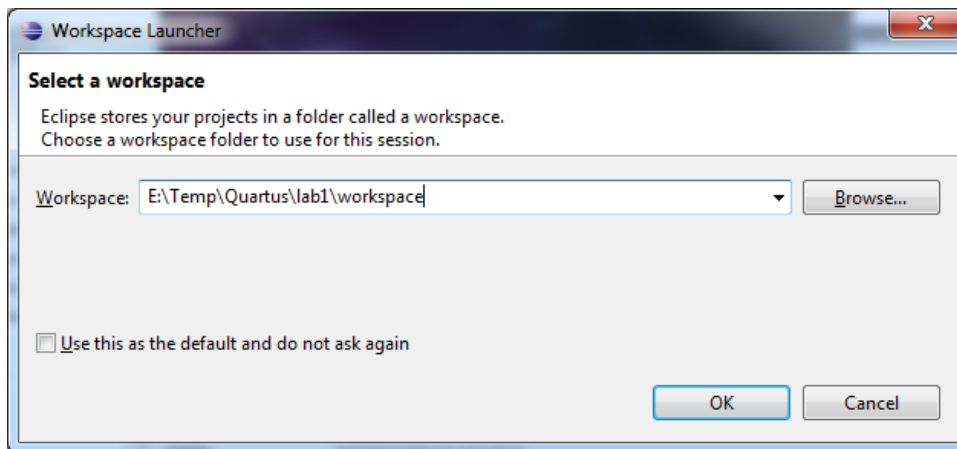


2.2. Lập trình phần mềm

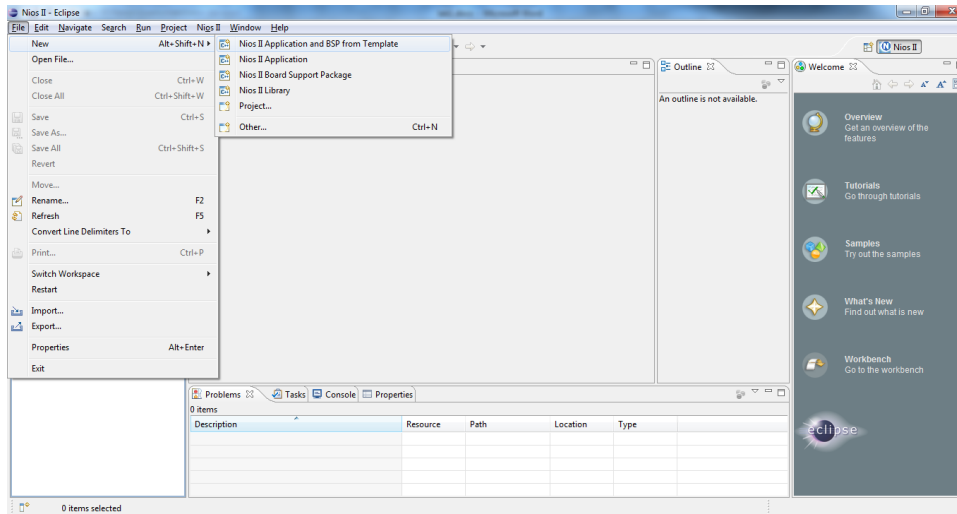
Khởi động chương trình “Nios II EDS” từ Qsys



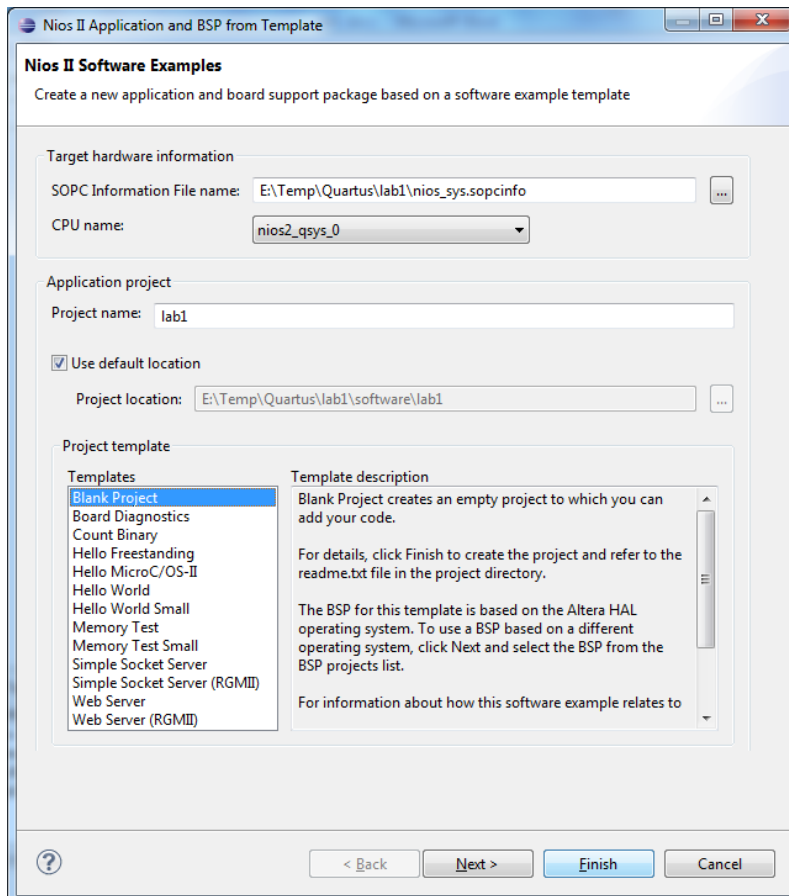
Chọn thư mục workspace, không được có khoảng trắng trong đường dẫn thư mục

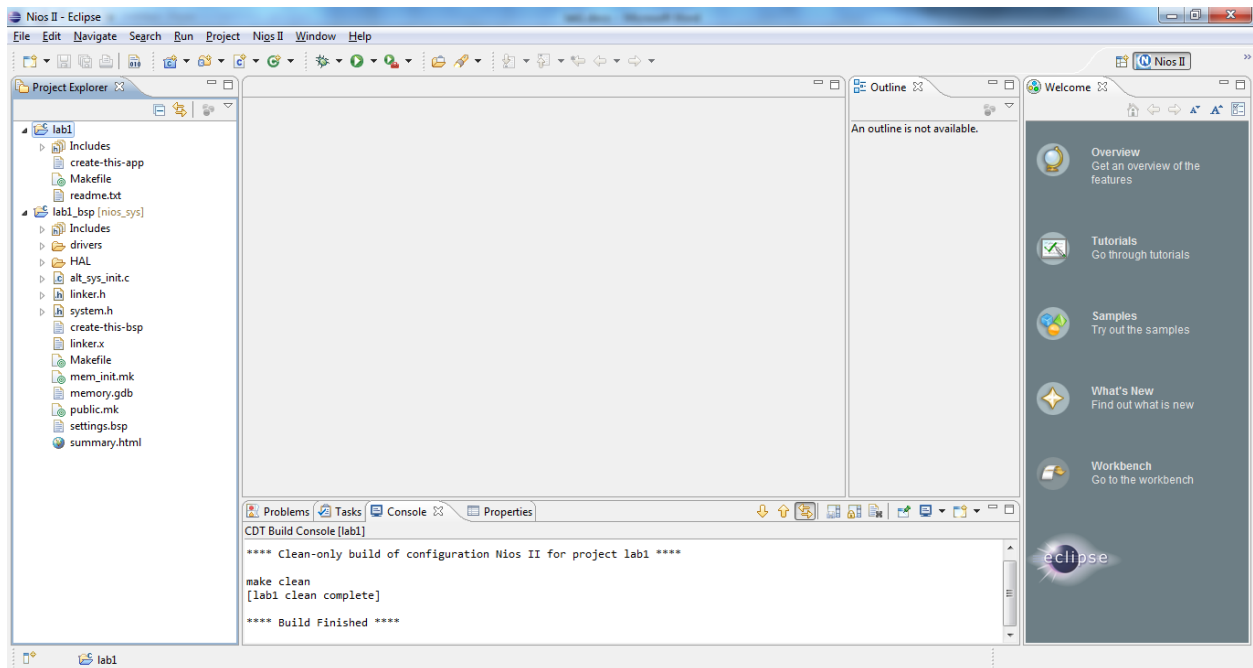


Tạo project “NIO S II EDS”

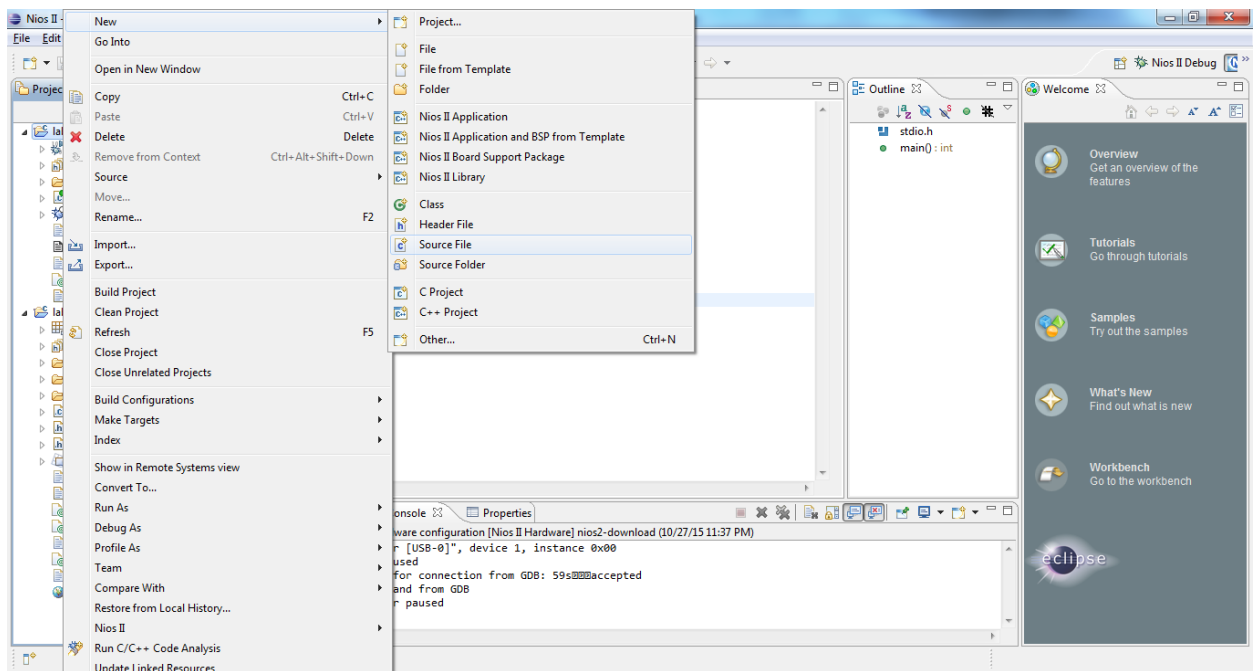


Ở mục “SOPC Information File name”, chọn file “nios_sys.sopcinfo”. Đặt tên project là “lab1”. Chọn “Templates” là “Blank project”. Click “Finish”

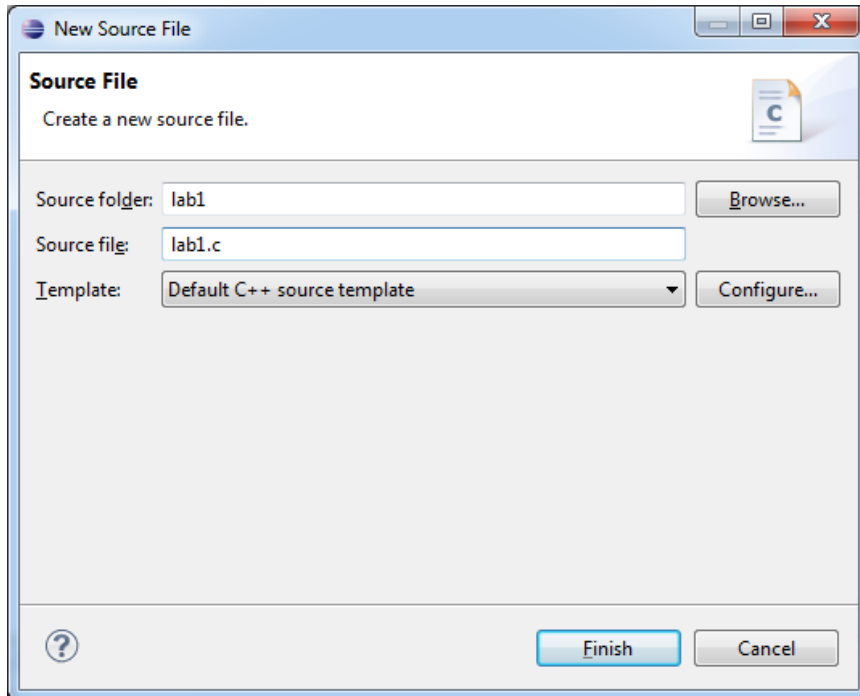




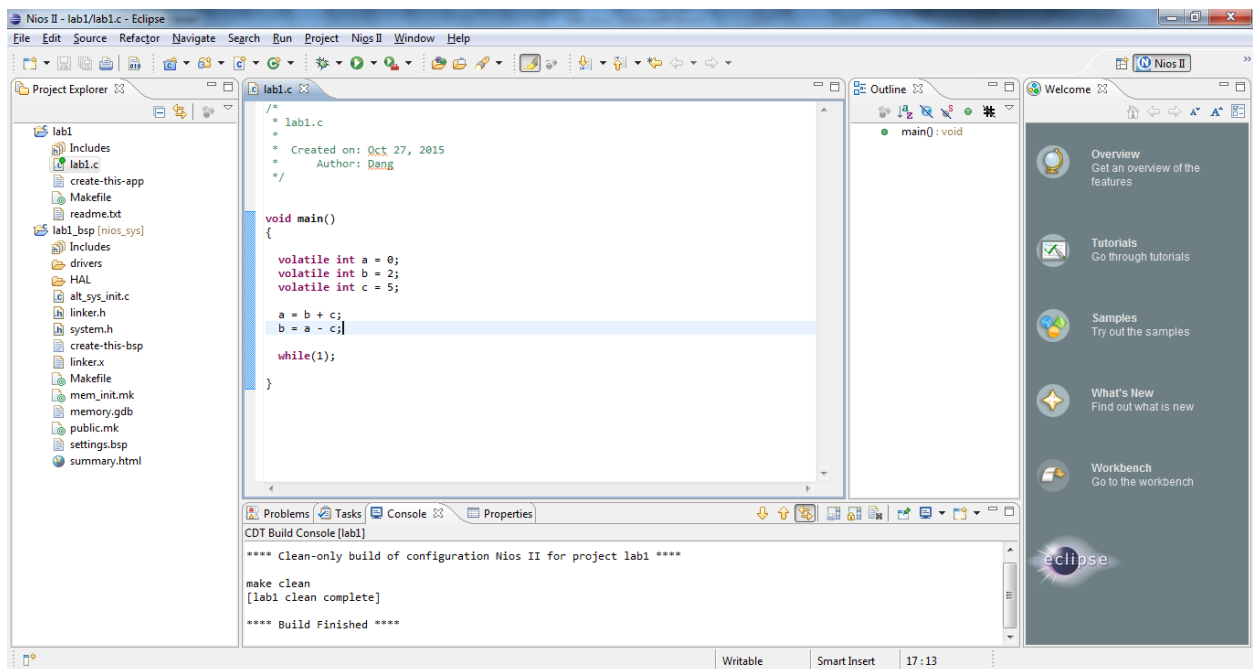
Click chuột phải vào lab1, chọn “New”, “Source File”



Đặt tên file là “lab1.c”, click “Finish”

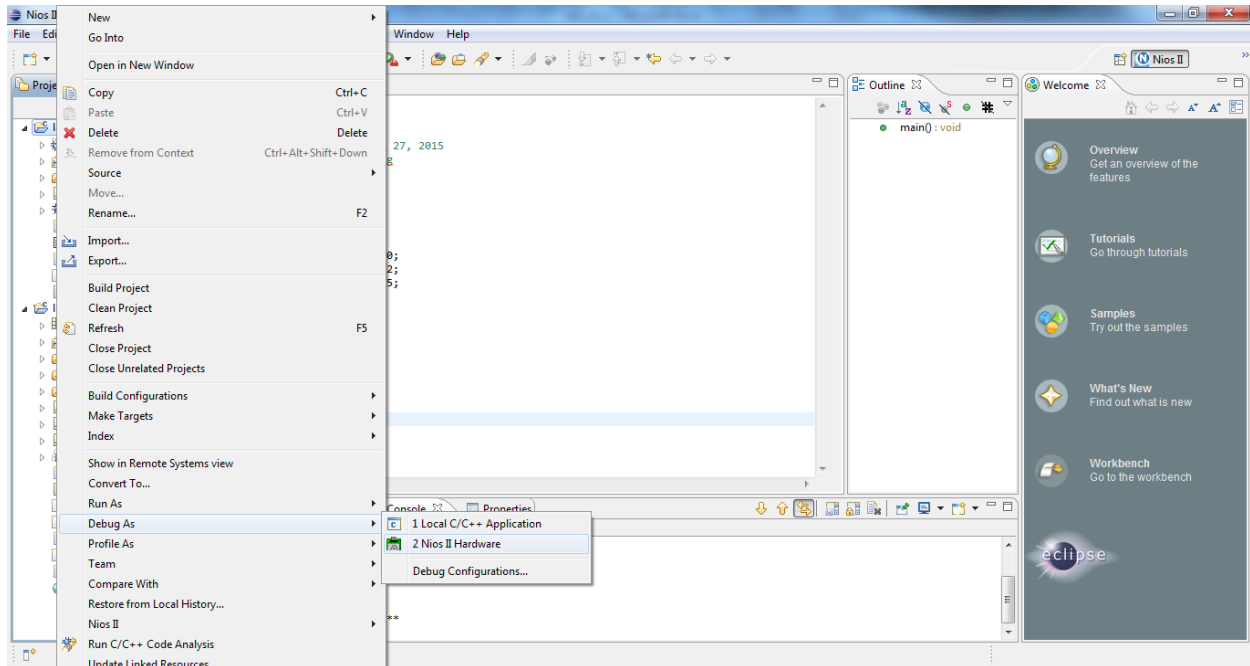


Nội dung của file “lab1.c” như bên dưới

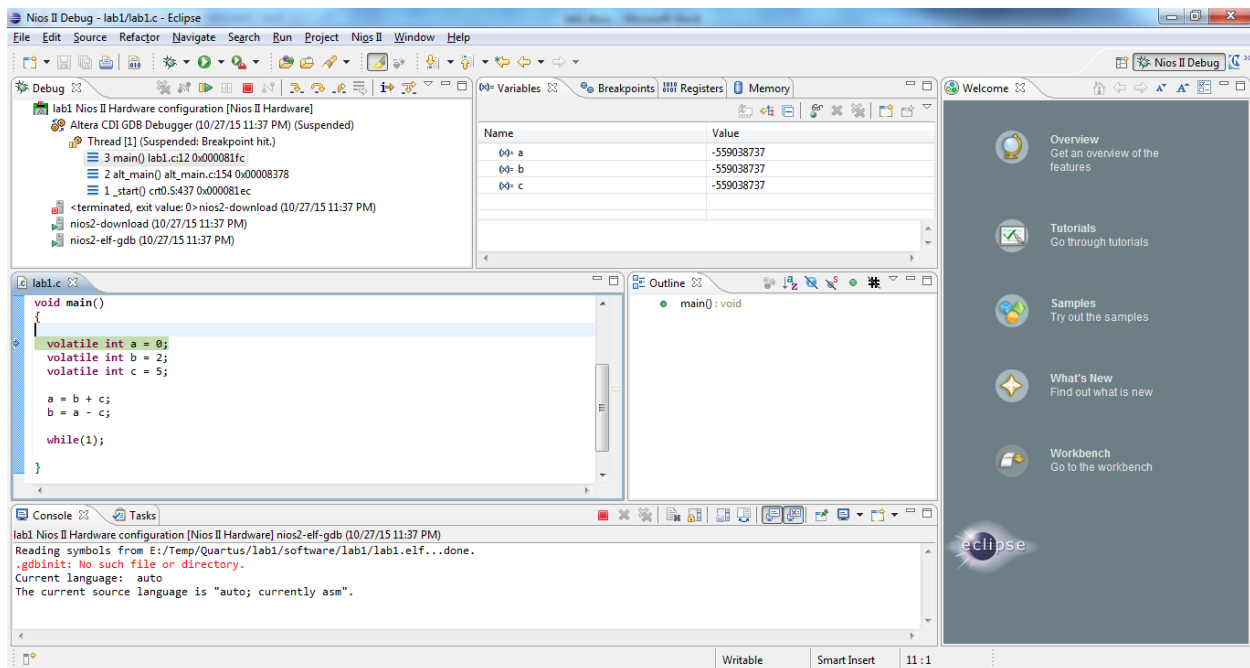


Vào menu “Project”, chọn “Build all”

Download phần mềm xuống board



Cửa sổ debug hiện ra như bên dưới



BÁO CÁO THỰC HÀNH

BÀI 1: GIỚI THIỆU

Sinh viên:

.....

Lớp: Nhóm:

Bài 1: Cho biết ý nghĩa và phím tắt của các biểu tượng sau trong cửa sổ debug của NIOS II EDS



Bài 2: Biên dịch và debug đoạn code sau

```
#include <stdio.h>

int main()
{
    printf("Hello from Nios II!\n");
    while(1);
}
```

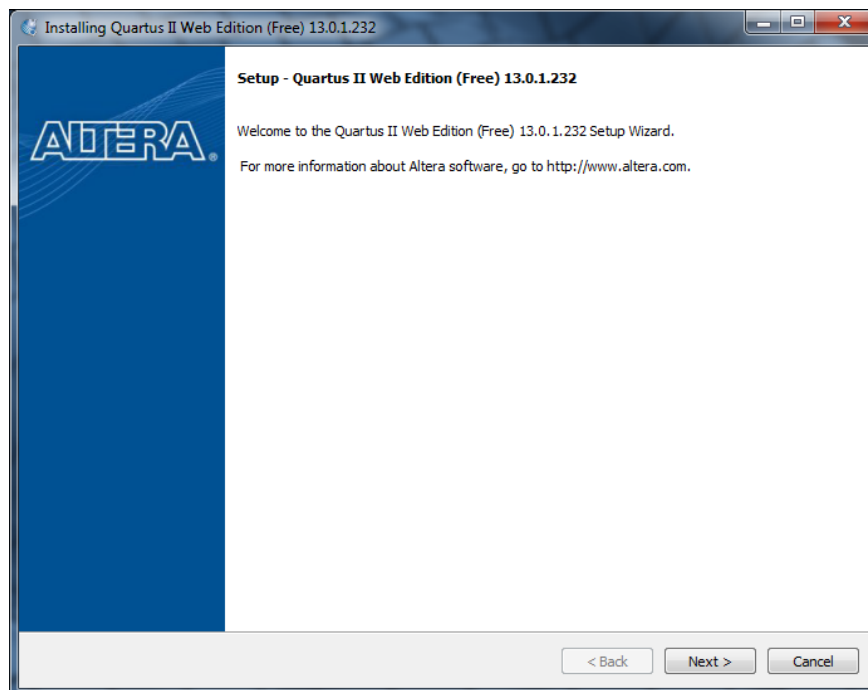
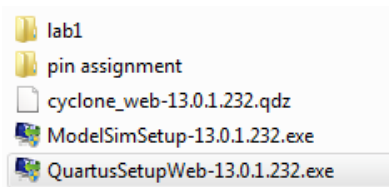
Phụ lục – Cài đặt Quartus II, Altera Modelsim, NIOS 2 IDE (version 13.0.1)

* Yêu cầu

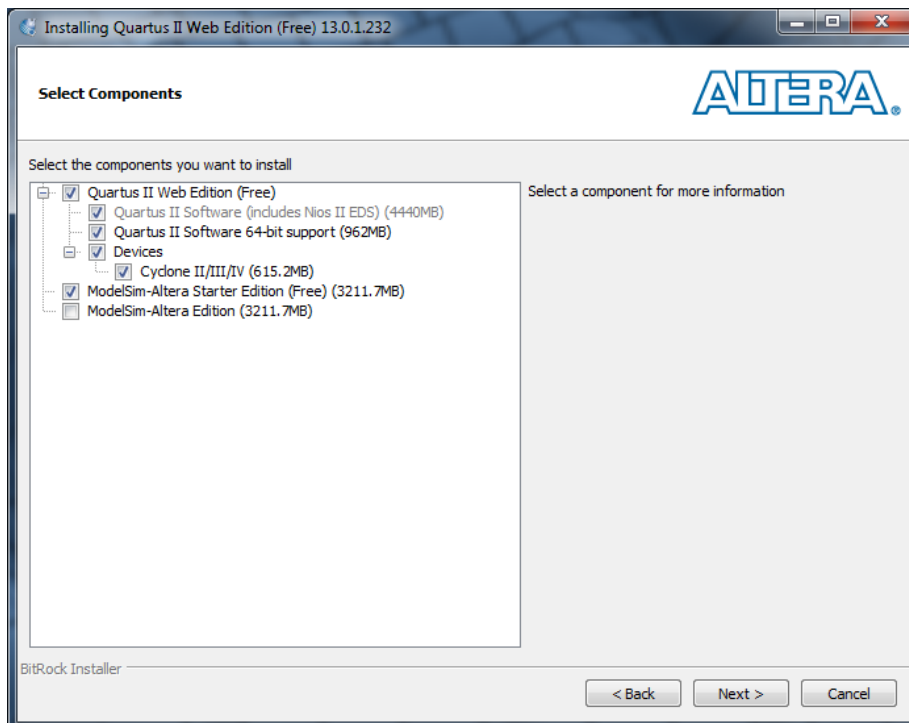
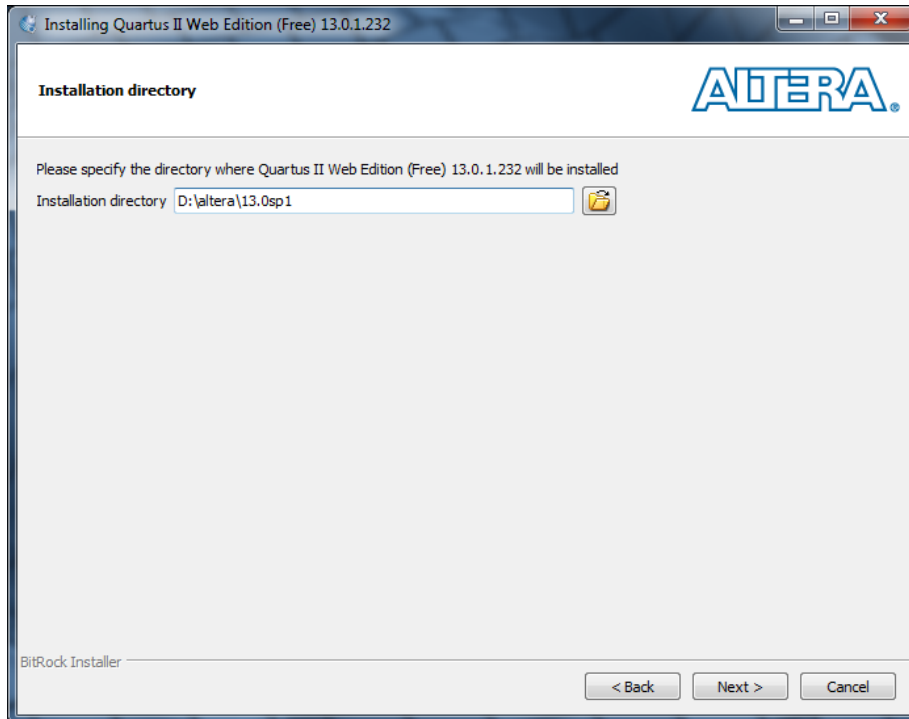
- Window 32-bit/64-bit
- Ổ cứng trống tối thiểu 10GB

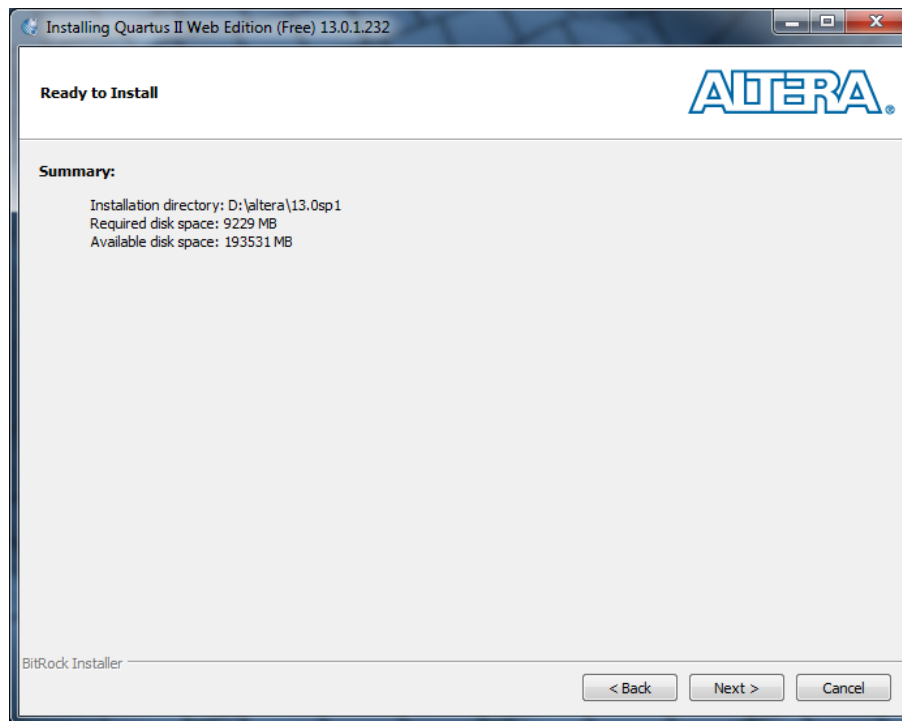
* Các bước cài đặt

Chạy file cài đặt “QuartusSetupWeb-13.0.1.232”



Đường dẫn thư mục cài đặt không được có khoảng trắng





Sau khi cài đặt xong, khởi động Quartus 2 và chọn như bên dưới

